

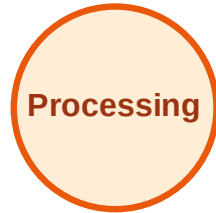
# BFS Traversal

Step 1: Start at A (Seeking target node K)

## Legend



Complete



Processing



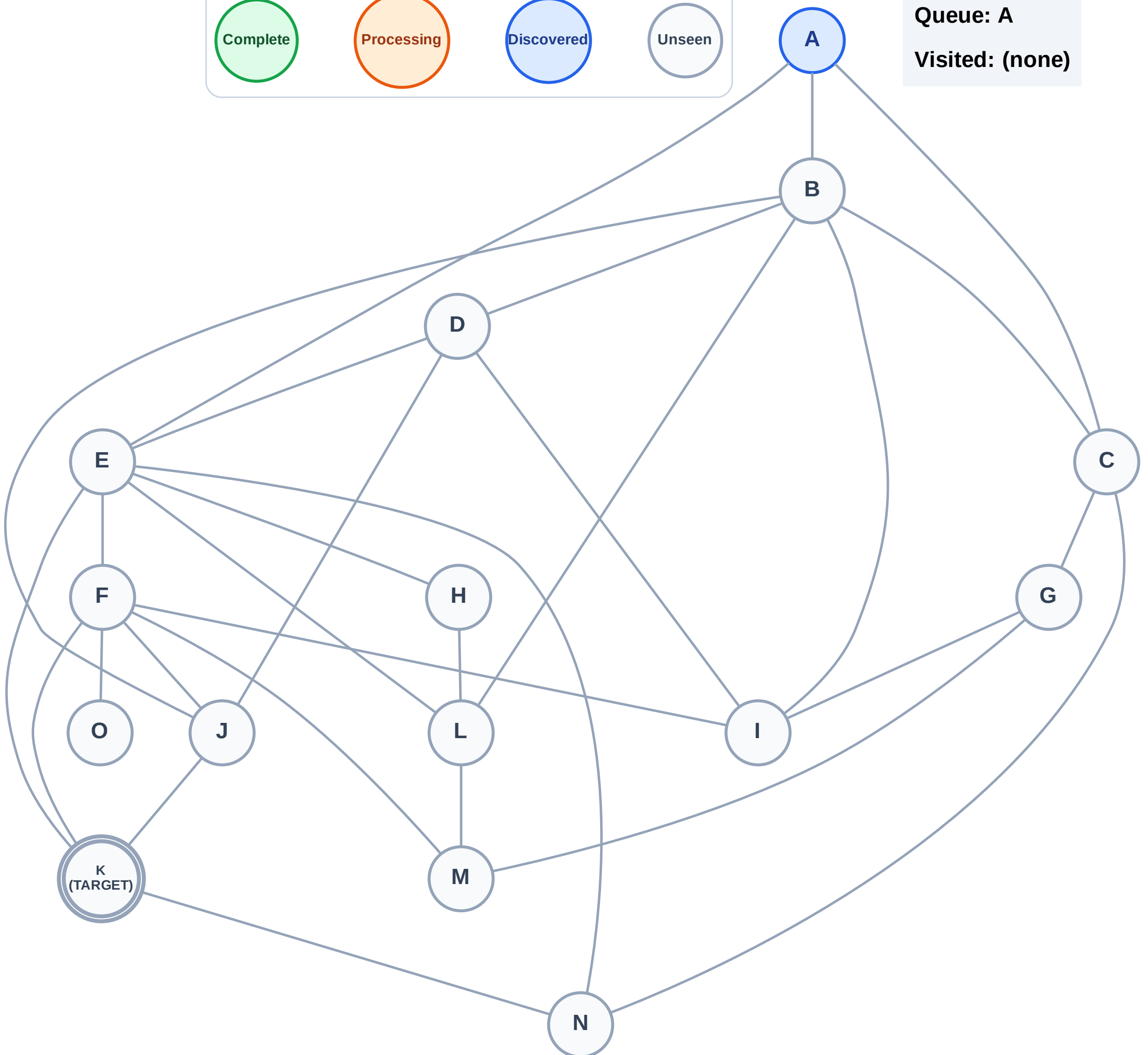
Discovered



Unseen

Queue: A

Visited: (none)



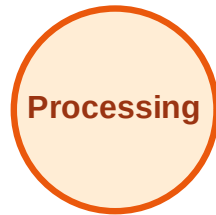
# BFS Traversal

Step 2: Process node A

## Legend



Complete



Processing



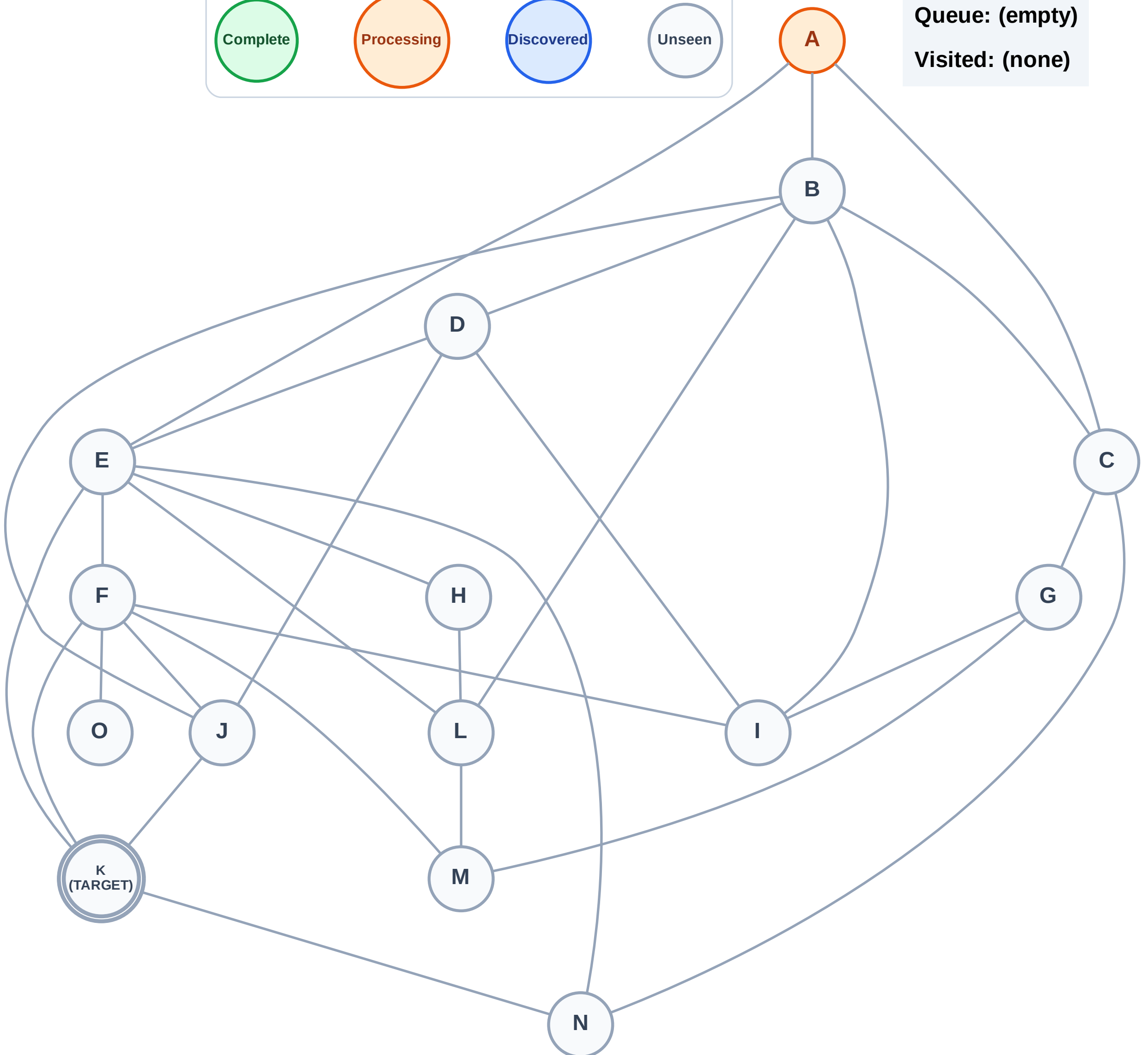
Discovered



Unseen

Queue: (empty)

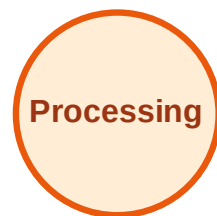
Visited: (none)



# BFS Traversal

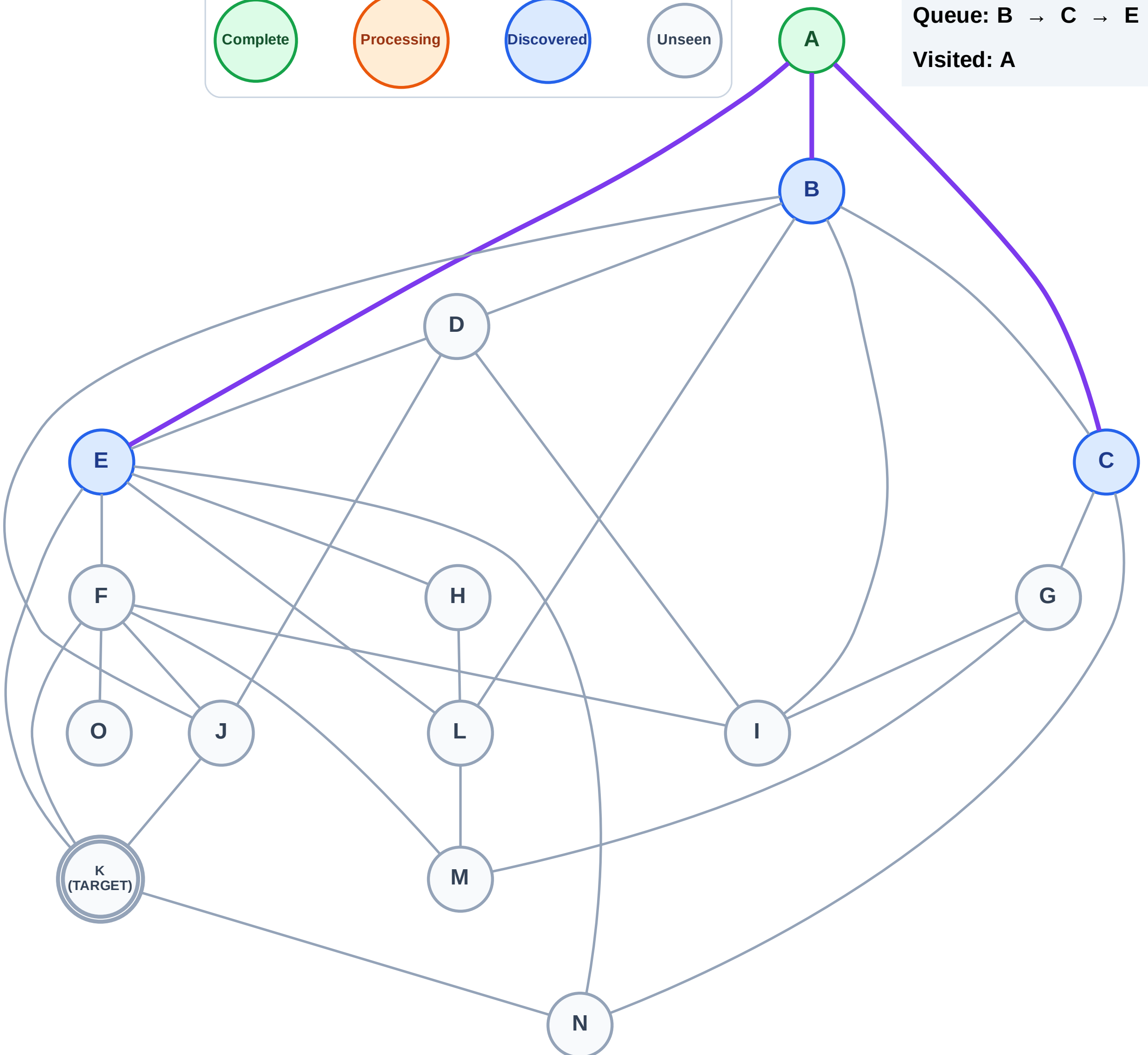
Step 3: Discover B, C, E from A

## Legend



Queue: B → C → E

Visited: A



### Step 4: Process node B

### Step 4: Process node B

**Legend**

The legend consists of four colored circles arranged horizontally. From left to right: a green circle labeled 'Complete', an orange circle labeled 'Processing', a blue circle labeled 'Discovered', and a grey circle labeled 'Unseen'.

**Legend**

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**Legend**

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**Legend**

The legend consists of four colored circles arranged horizontally. From left to right: a green circle labeled 'Complete', an orange circle labeled 'Processing', a blue circle labeled 'Discovered', and a grey circle labeled 'Unseen'.

**Legend**

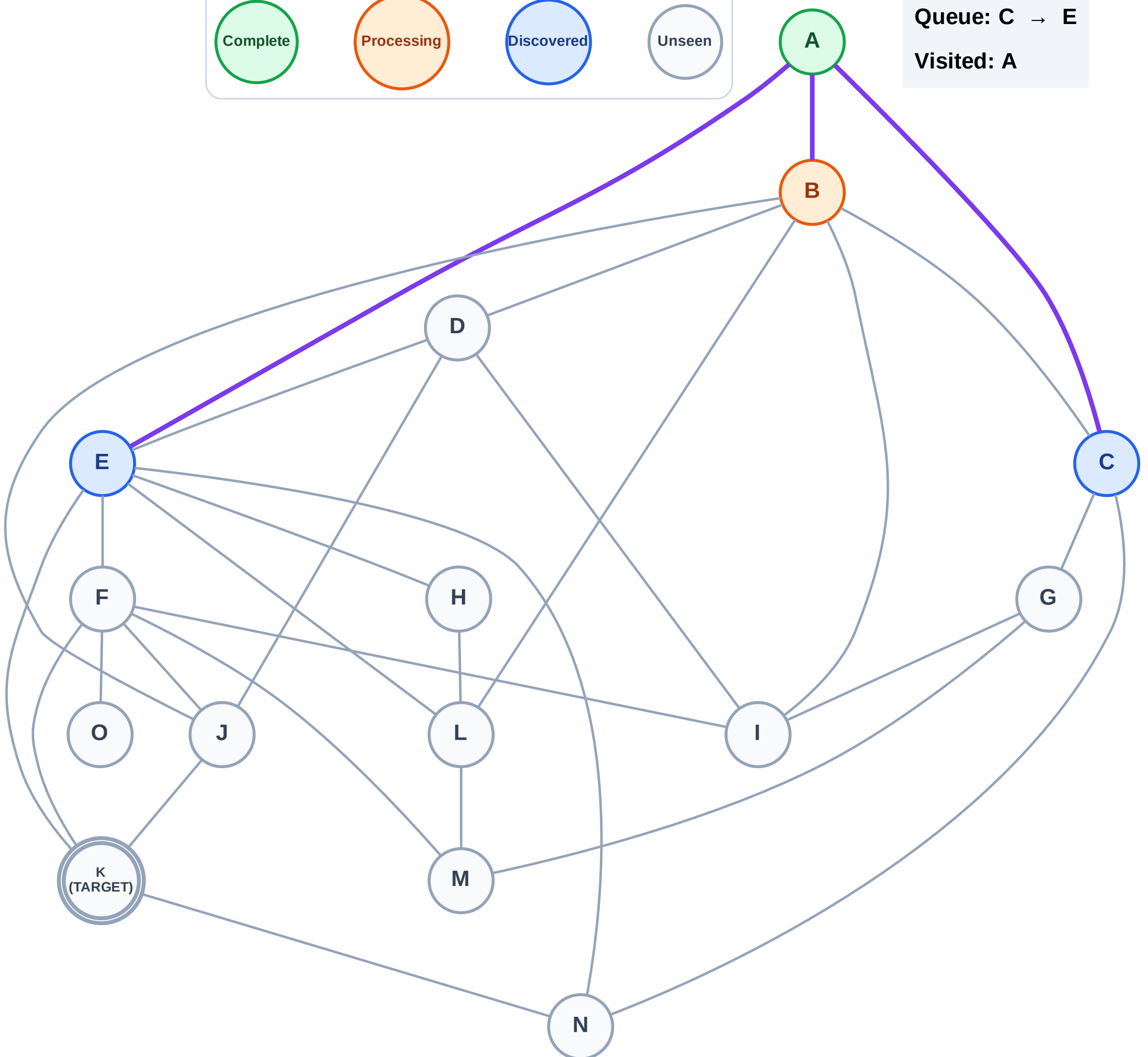
The legend consists of four colored circles arranged horizontally. From left to right: a green circle labeled 'Complete', an orange circle labeled 'Processing', a blue circle labeled 'Discovered', and a grey circle labeled 'Unseen'.

Queue: C → E

Visited: A

Queue: C → E

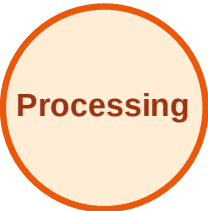
Visited: A



# BFS Traversal

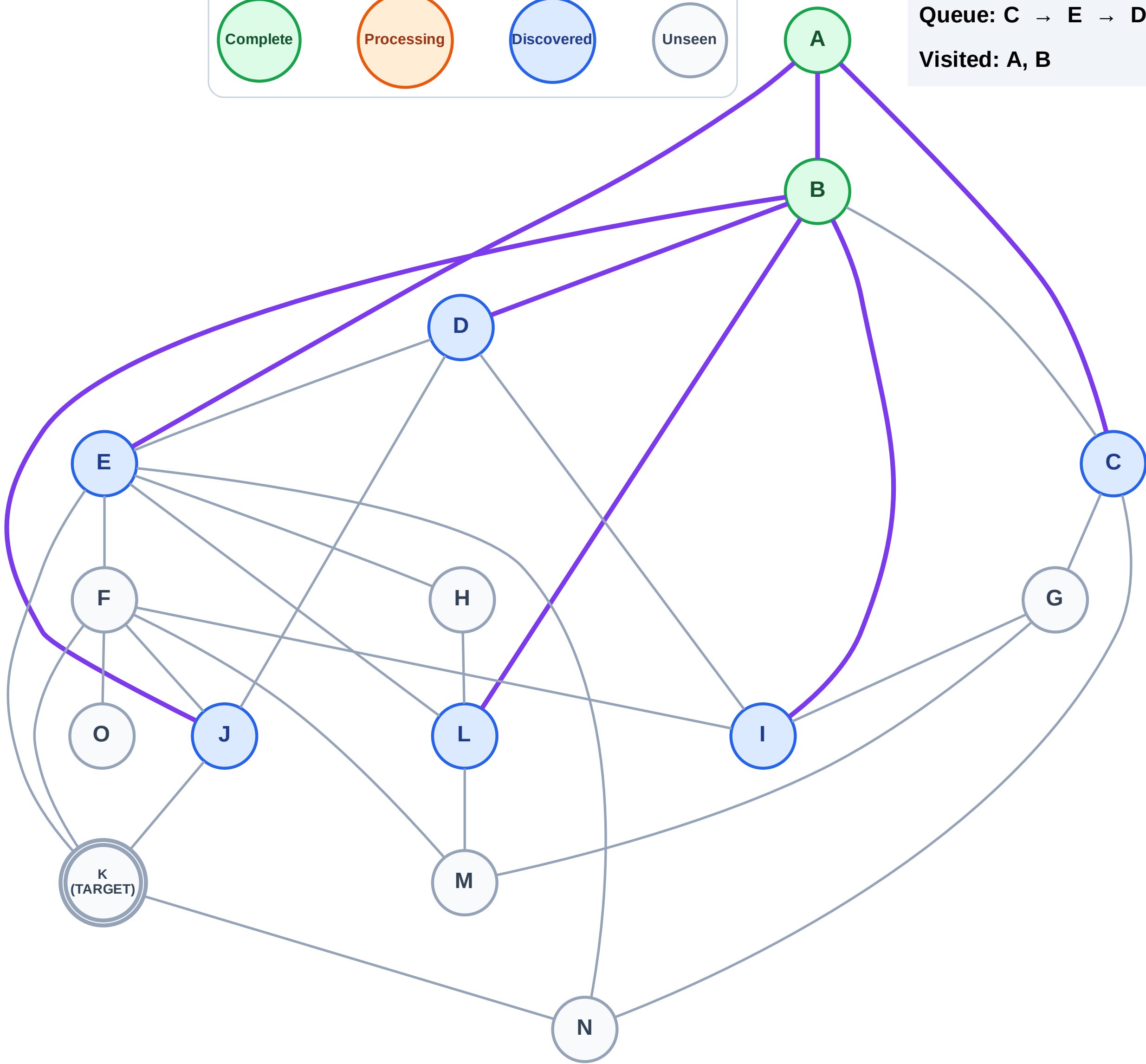
Step 5: Discover D, I, J, L from B

## Legend



Queue: C → E → D → I → J → L

Visited: A, B



# BFS Traversal

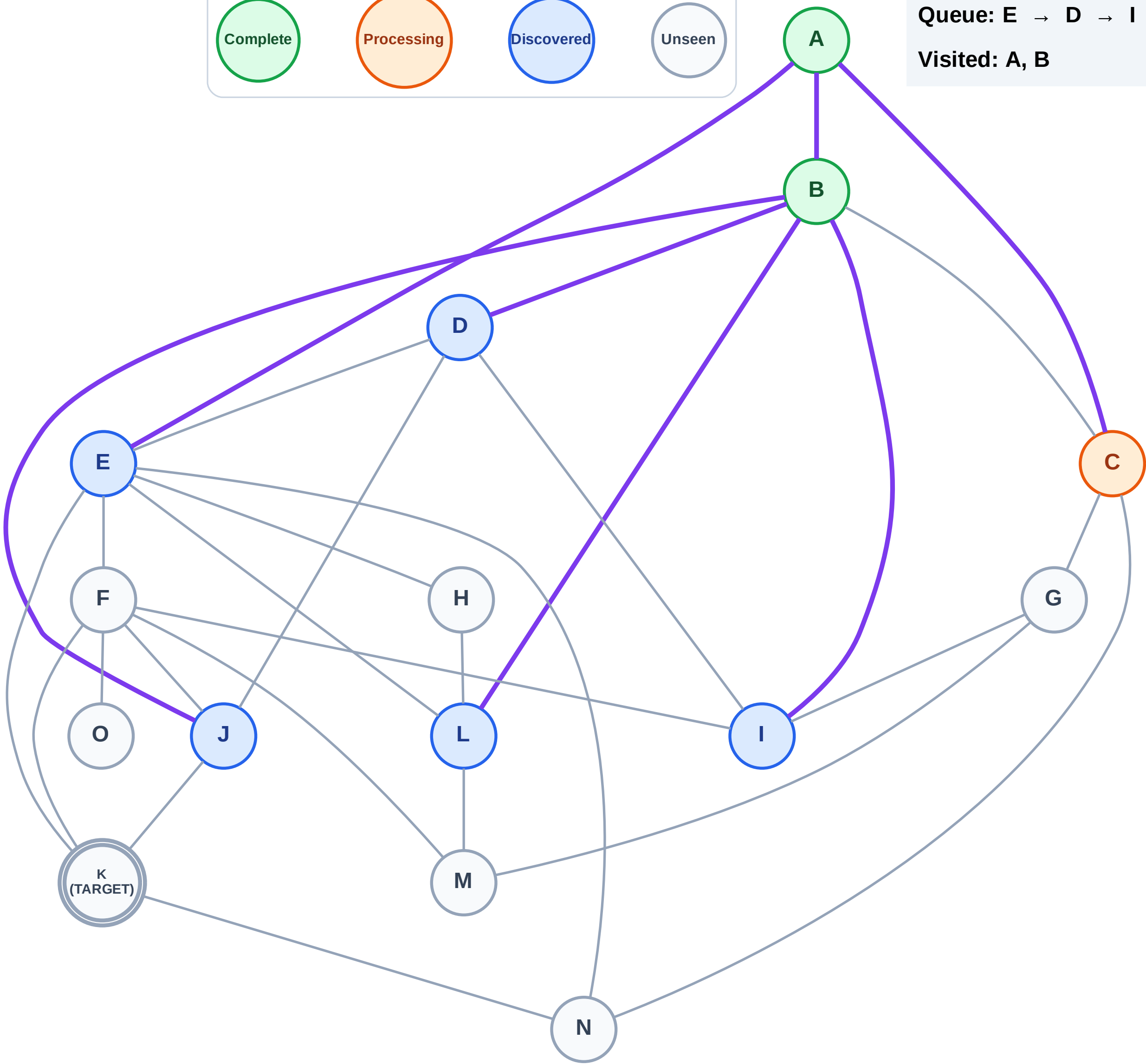
Step 6: Process node C

## Legend



Queue: E → D → I → J → L

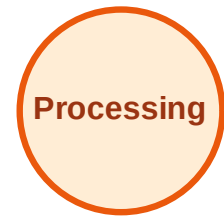
Visited: A, B



# BFS Traversal

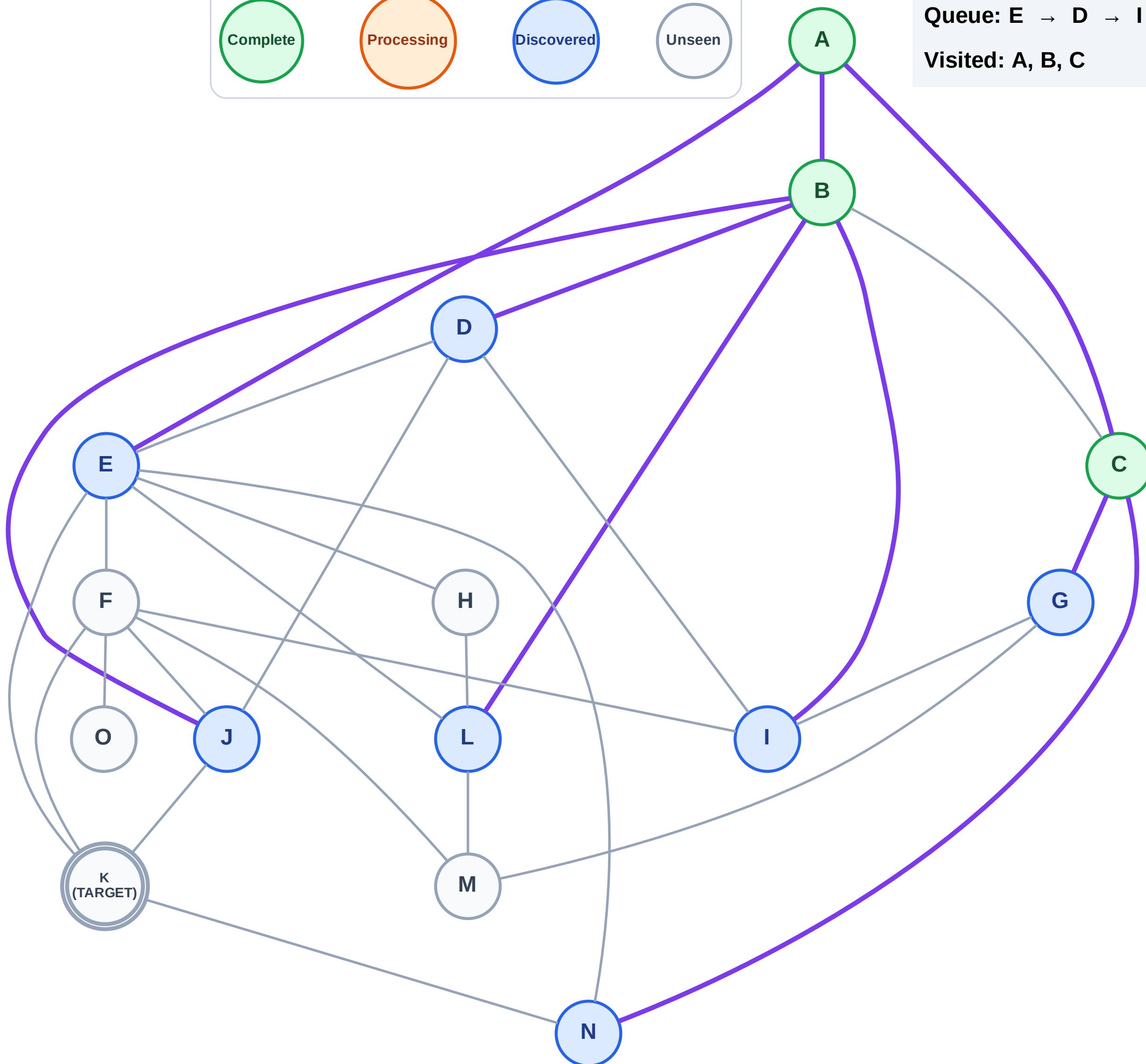
Step 7: Discover G, N from C

## Legend



Queue: E → D → I → J → L → G → N

Visited: A, B, C



# BFS Traversal

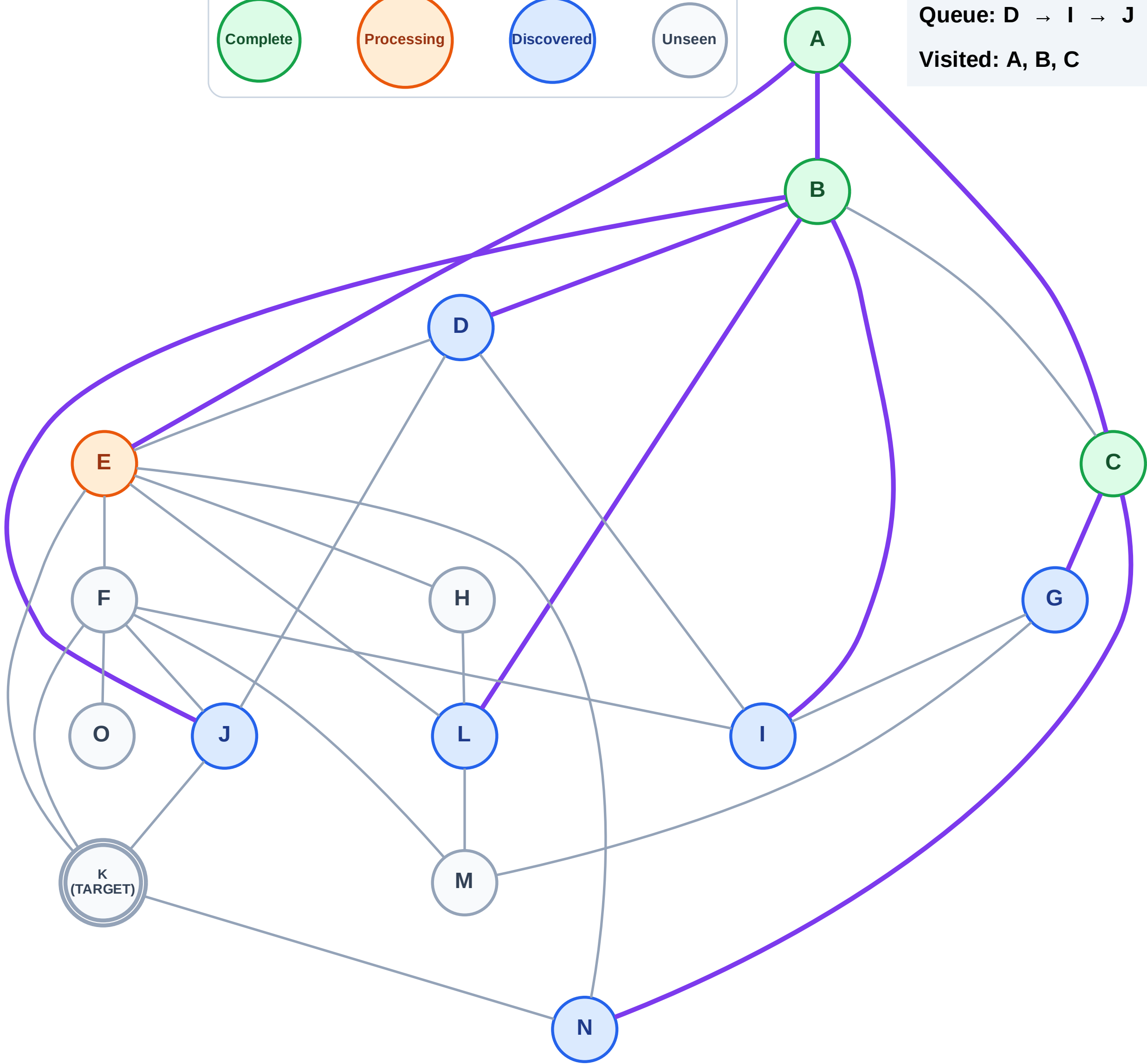
Step 8: Process node E

## Legend



Queue: D → I → J → L → G → N

Visited: A, B, C



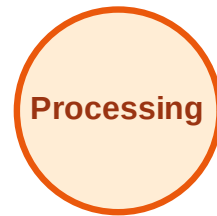
# BFS Traversal

Step 9: Discover F, H, K from E

## Legend



Complete



Processing



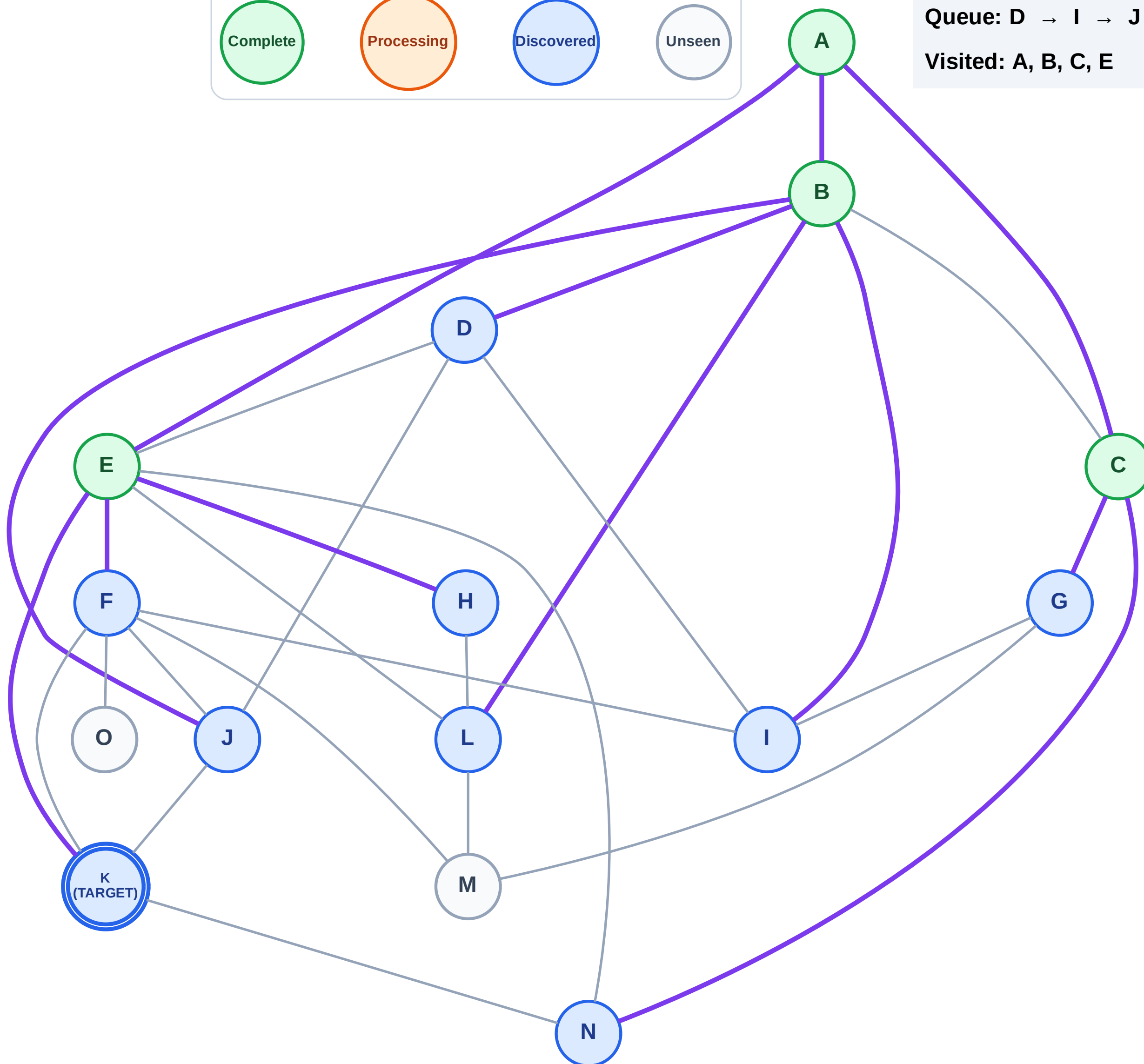
Discovered



Unseen

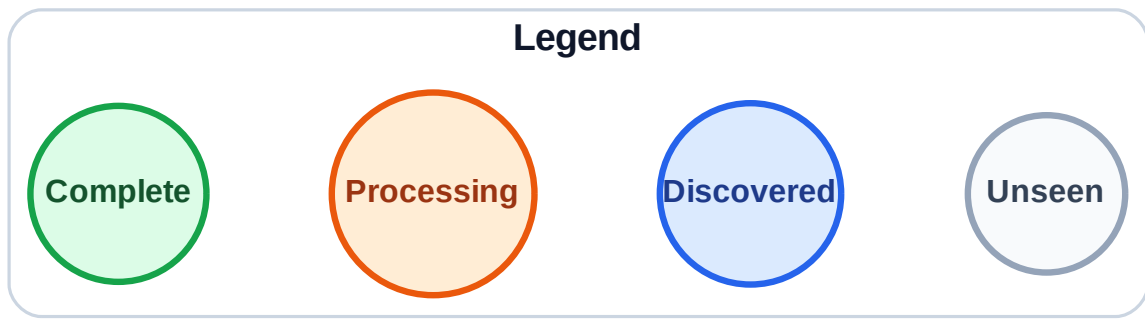
Queue: D → I → J → L → G → N → F → H → K

Visited: A, B, C, E



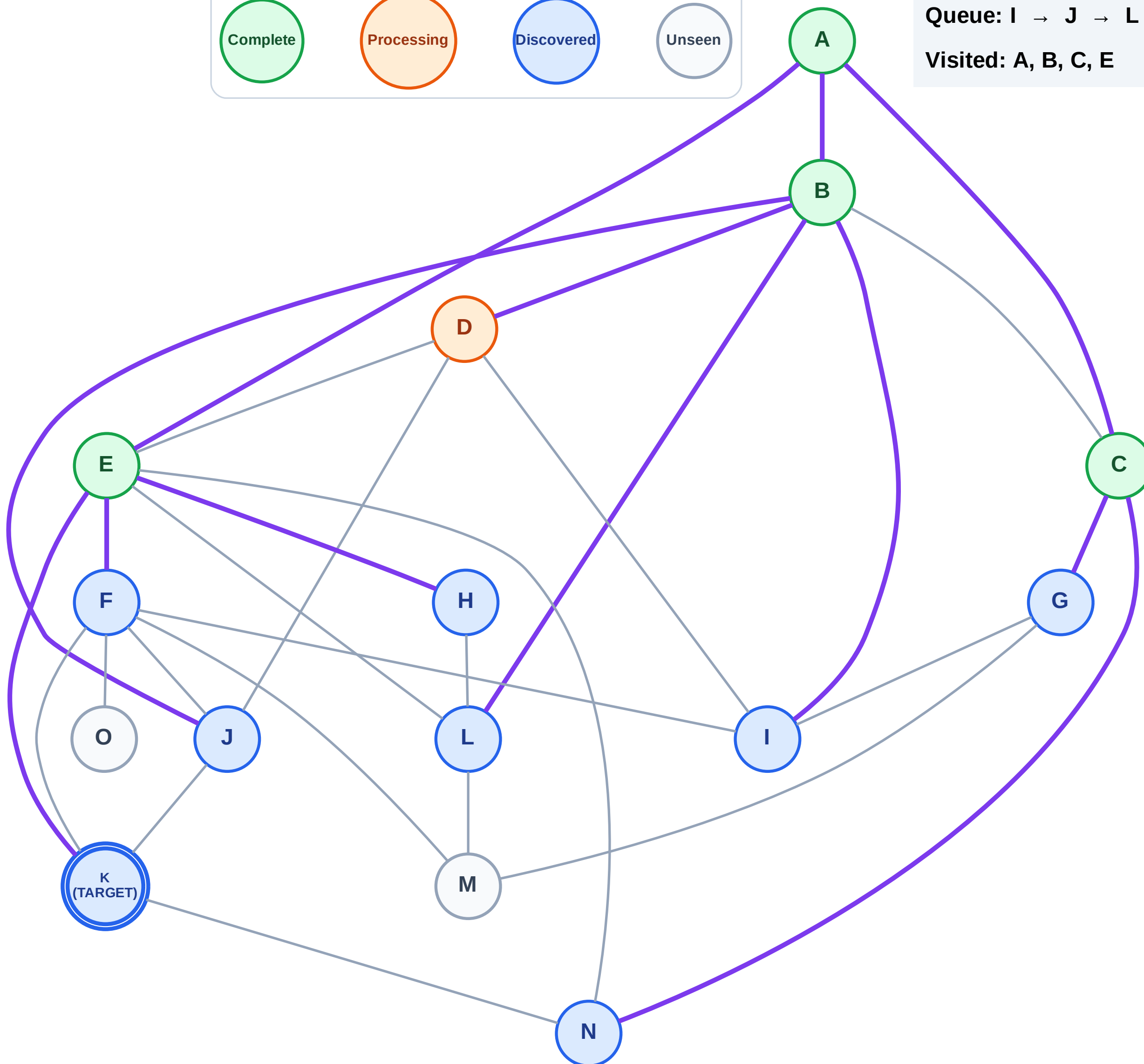
# BFS Traversal

Step 10: Process node D



Queue: I → J → L → G → N → F → H → K

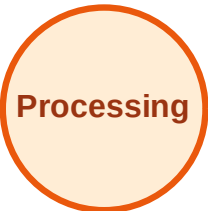
Visited: A, B, C, E



# BFS Traversal

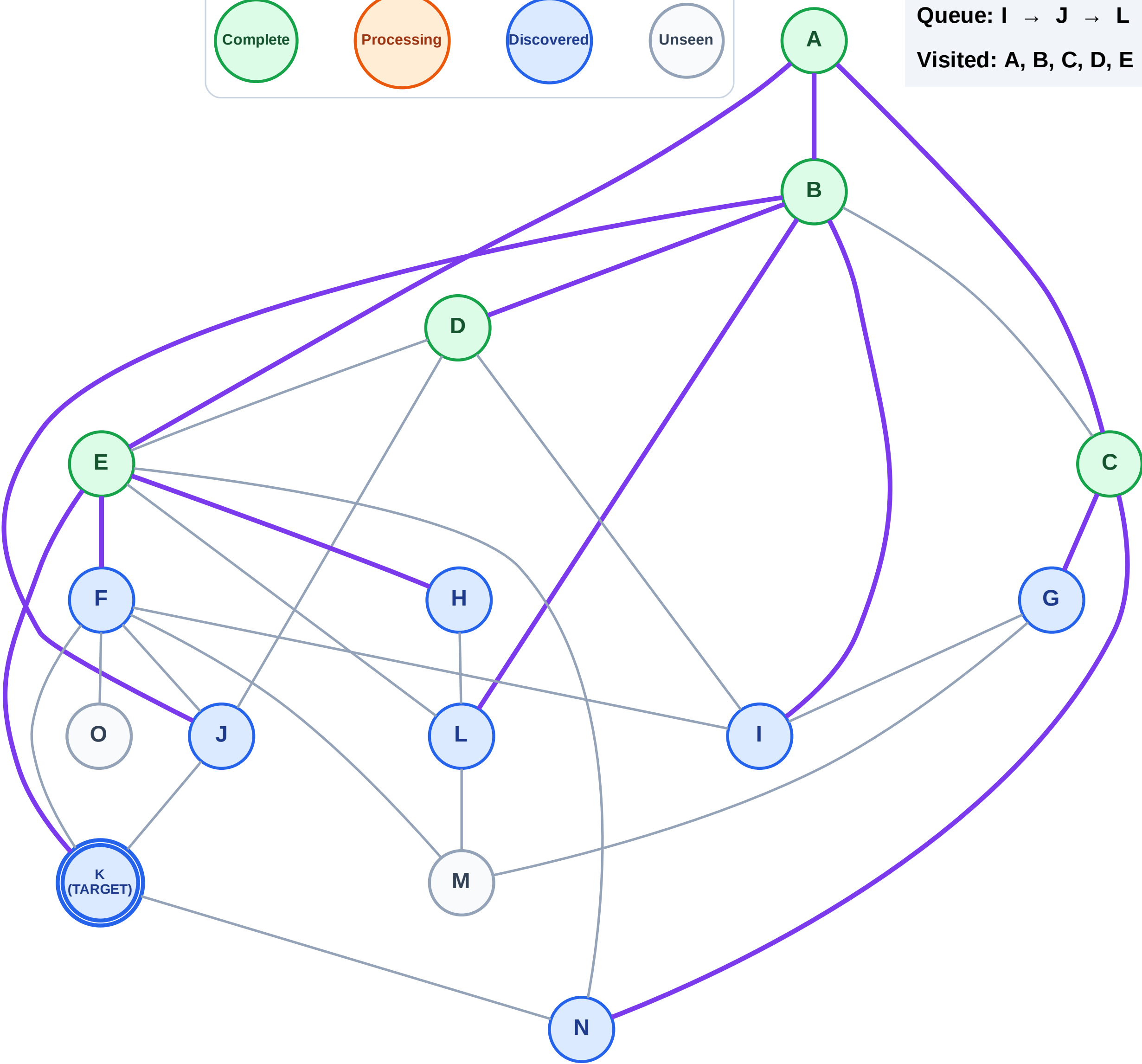
Step 11: No new neighbors from D

## Legend



Queue: I → J → L → G → N → F → H → K

Visited: A, B, C, D, E



# BFS Traversal

Step 12: Process node I

Legend

Complete

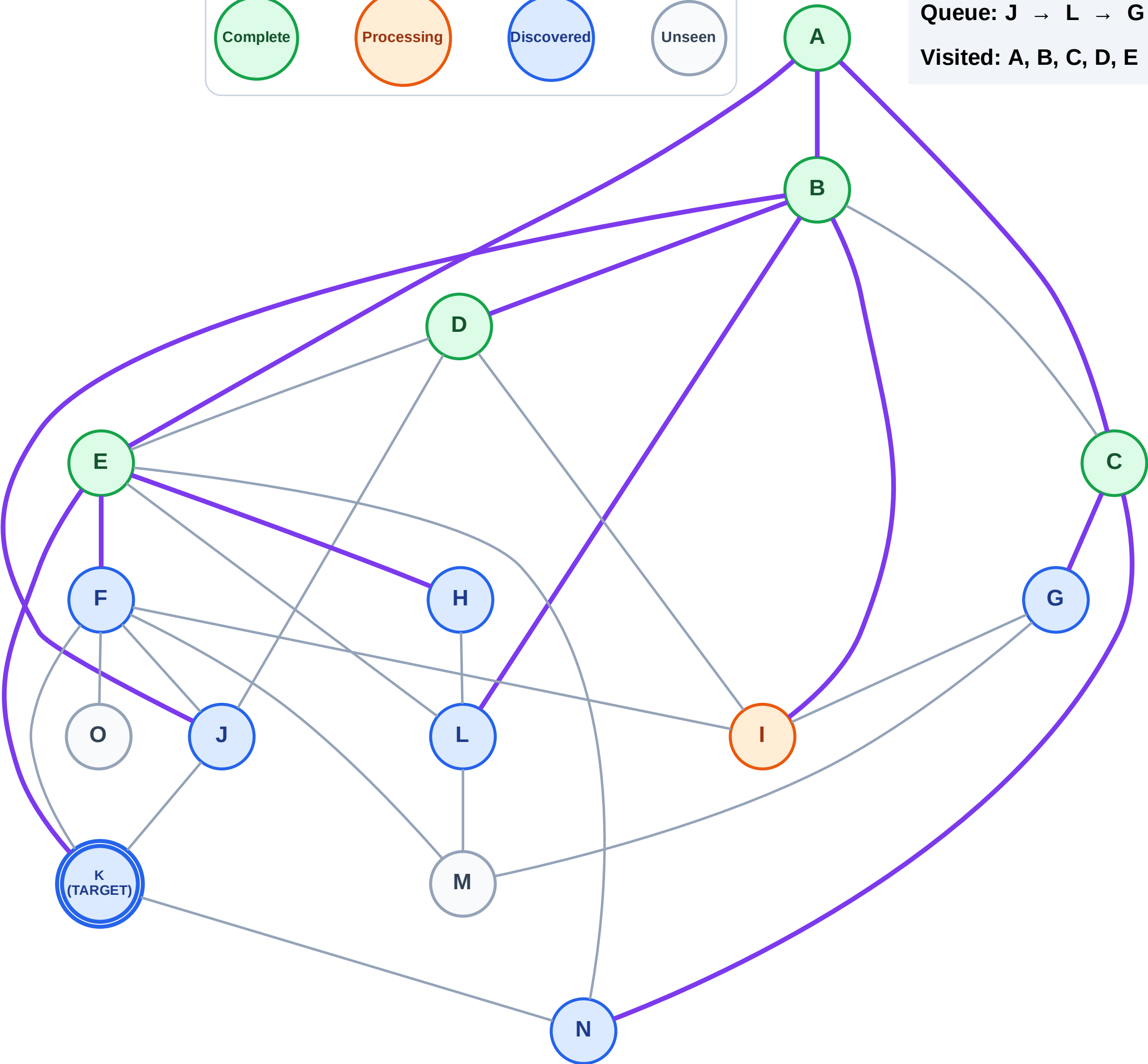
Processing

Discovered

Unseen

Queue: J → L → G → N → F → H → K

Visited: A, B, C, D, E



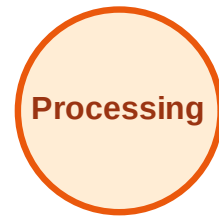
# BFS Traversal

Step 13: No new neighbors from I

## Legend



Complete



Processing



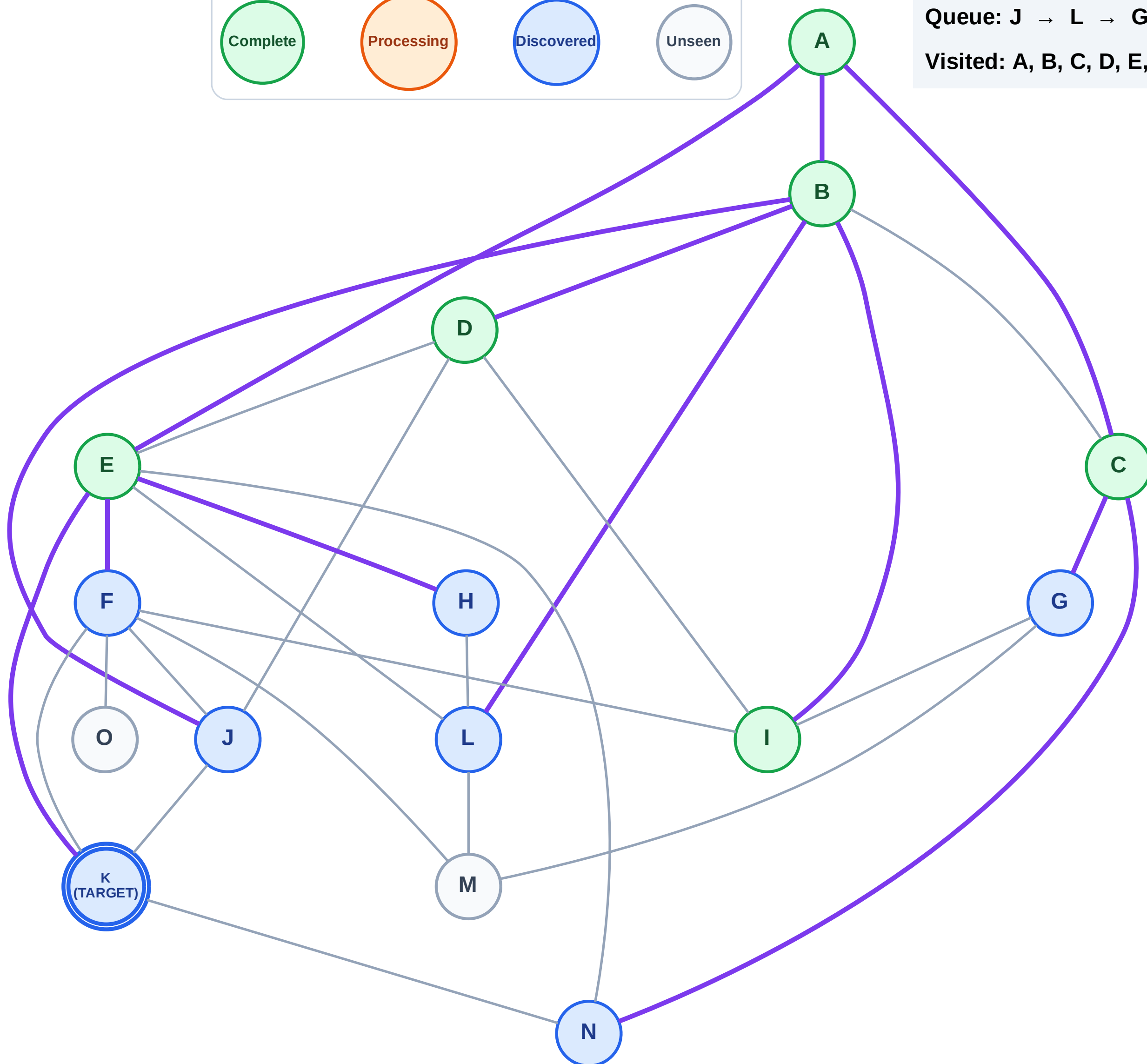
Discovered



Unseen

Queue: J → L → G → N → F → H → K

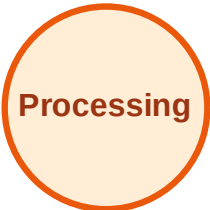
Visited: A, B, C, D, E, I



# BFS Traversal

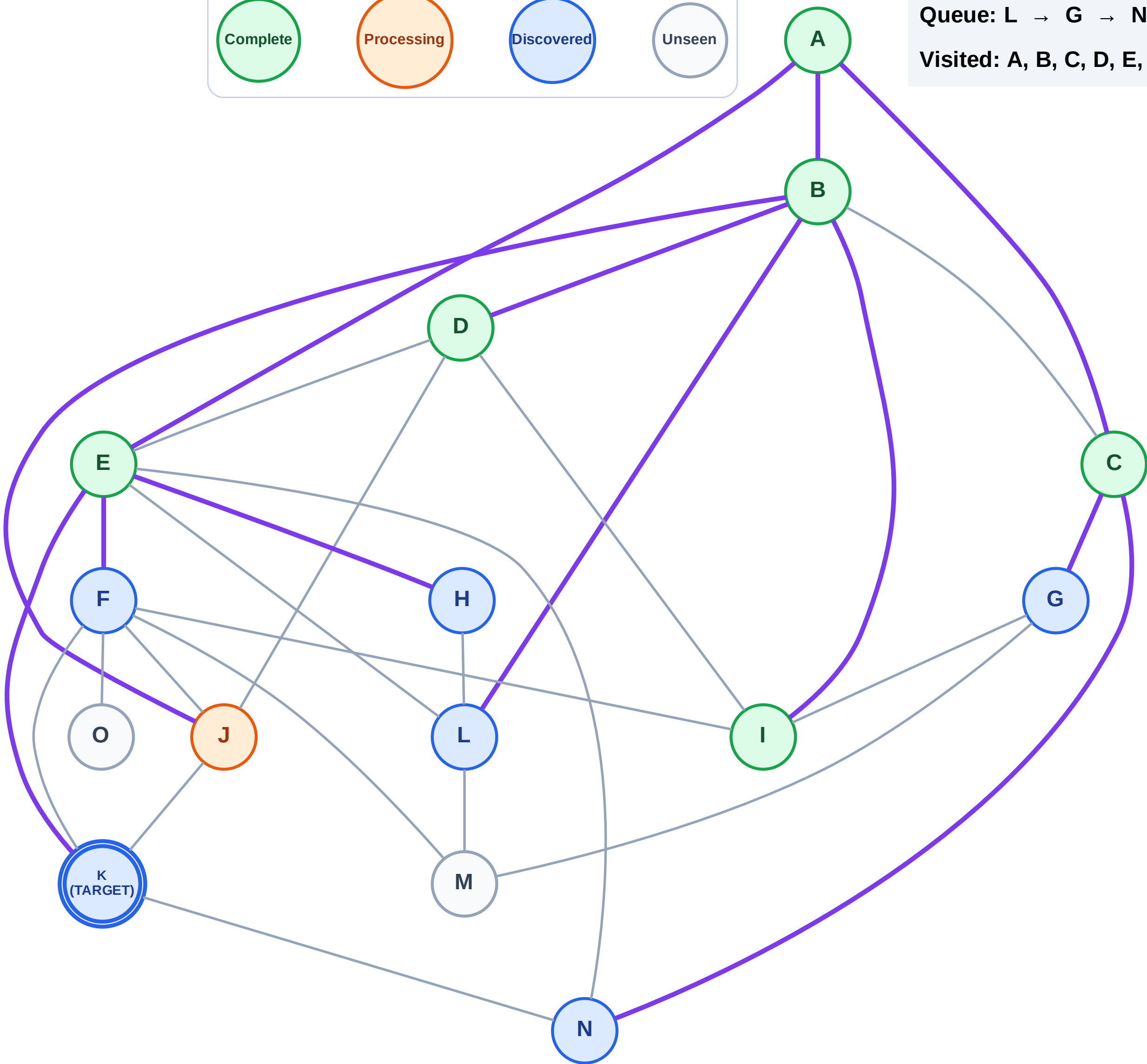
Step 14: Process node J

## Legend



Queue: L → G → N → F → H → K

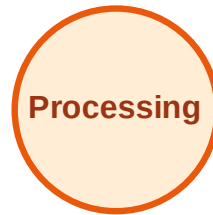
Visited: A, B, C, D, E, I



# BFS Traversal

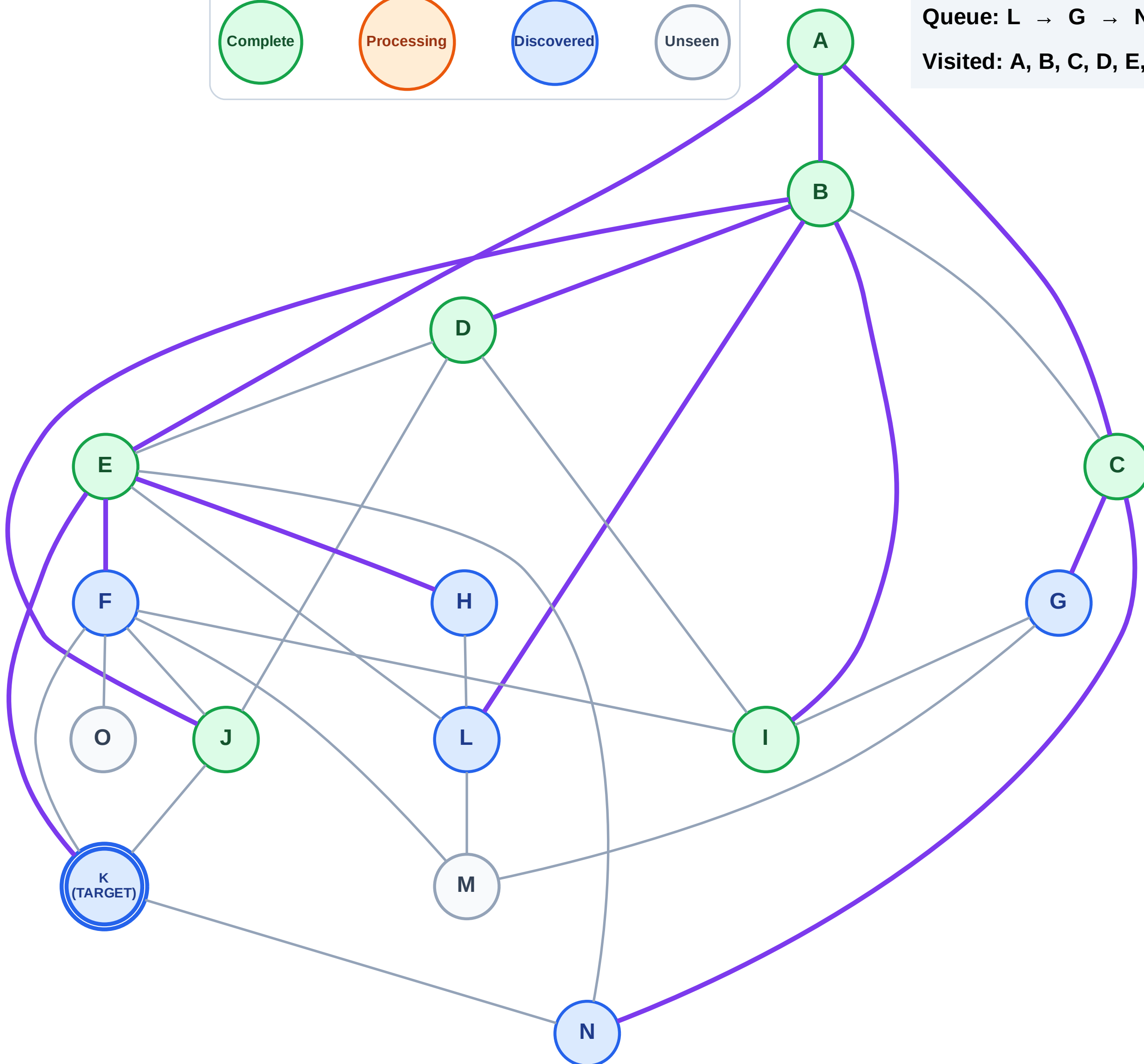
Step 15: No new neighbors from J

## Legend



Queue: L → G → N → F → H → K

Visited: A, B, C, D, E, I, J



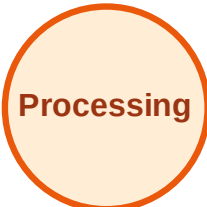
# BFS Traversal

Step 16: Process node L

## Legend



Complete



Processing



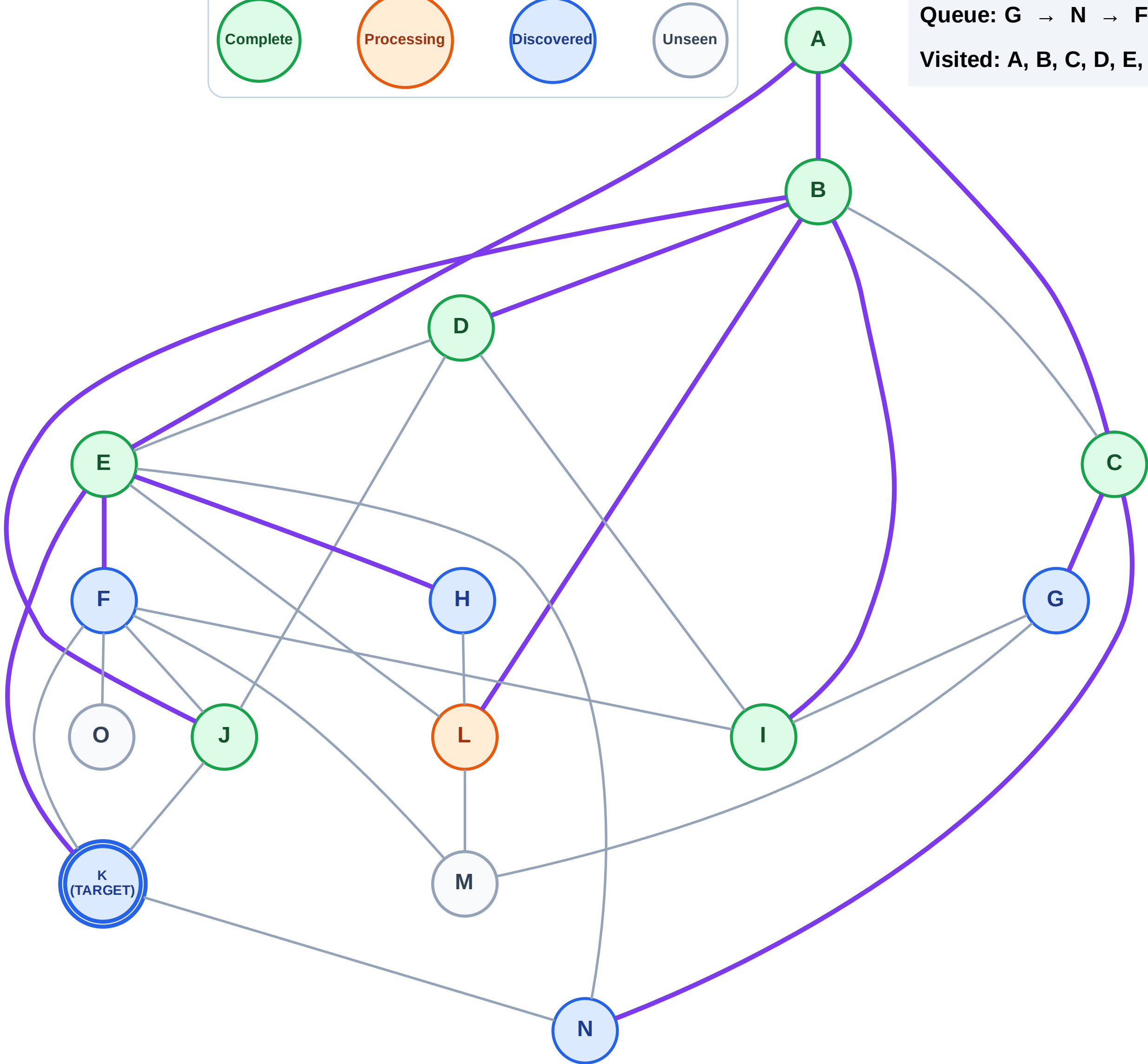
Discovered



Unseen

Queue: G → N → F → H → K

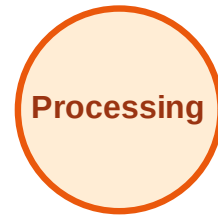
Visited: A, B, C, D, E, I, J



# BFS Traversal

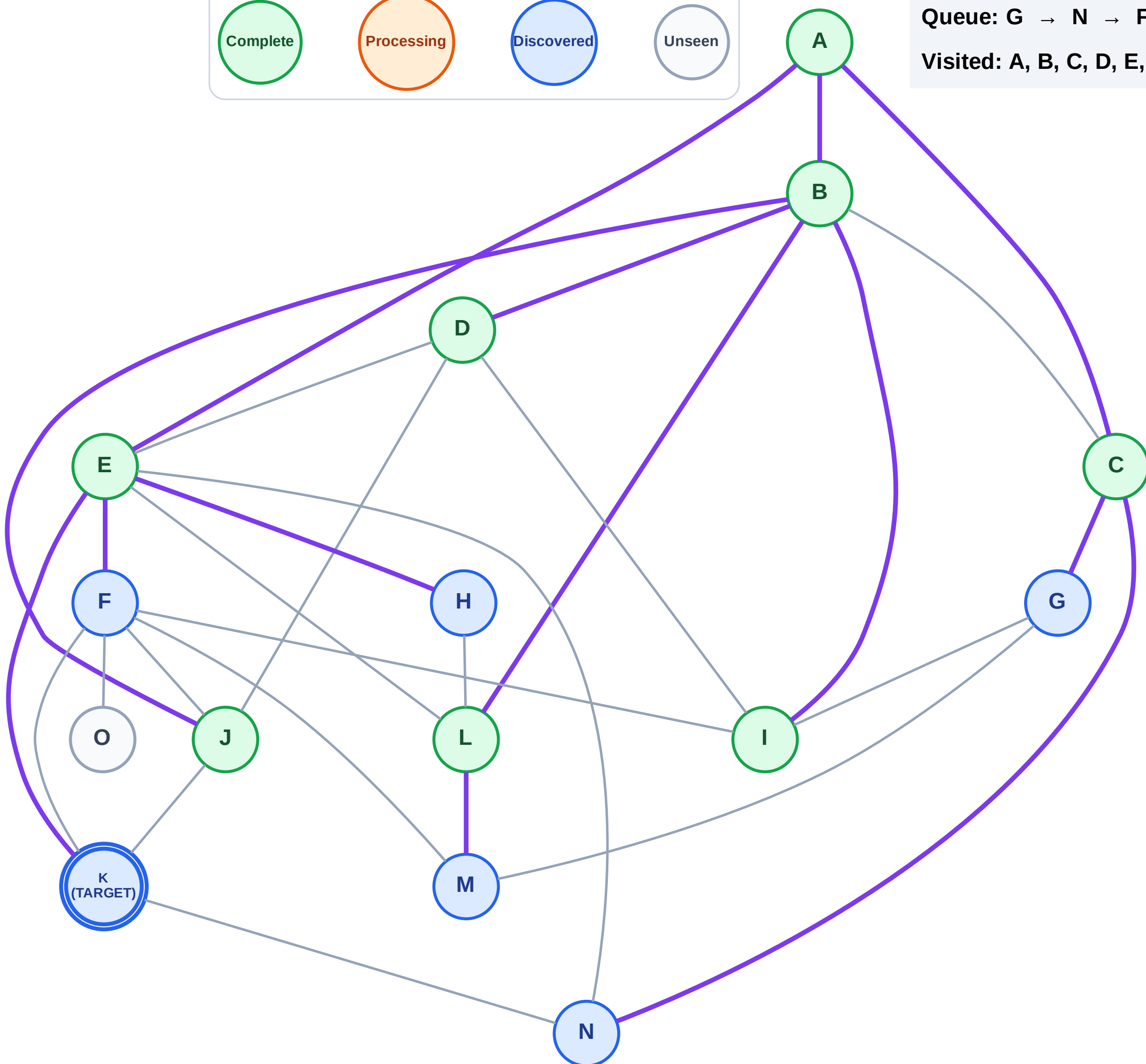
Step 17: Discover M from L

## Legend



Queue: G → N → F → H → K → M

Visited: A, B, C, D, E, I, J, L



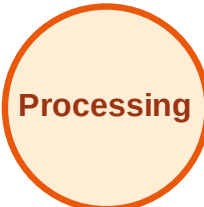
# BFS Traversal

Step 18: Process node G

## Legend



Complete



Processing



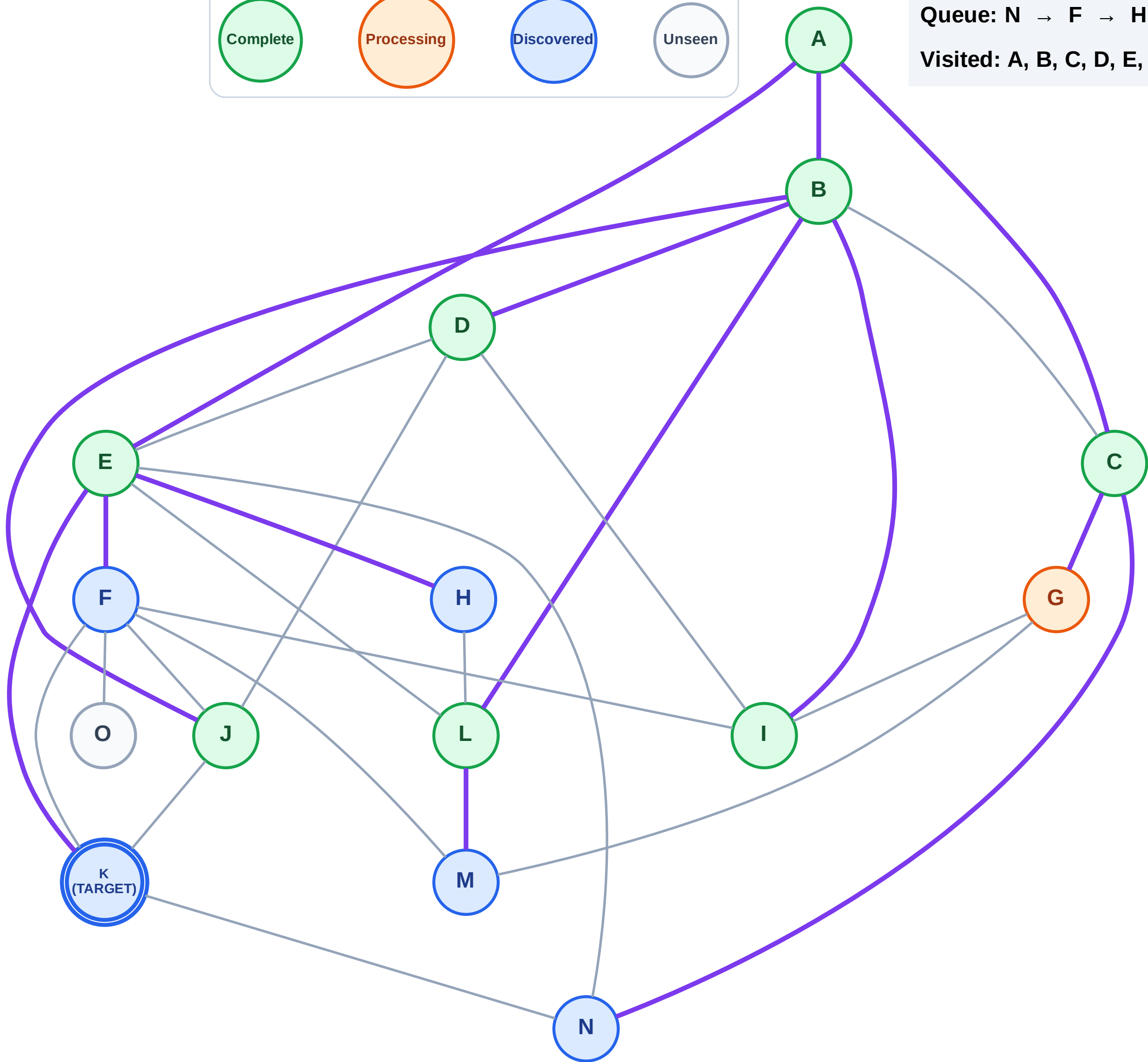
Discovered



Unseen

Queue: N → F → H → K → M

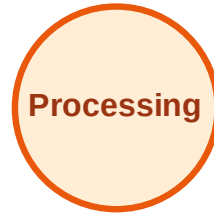
Visited: A, B, C, D, E, I, J, L



# BFS Traversal

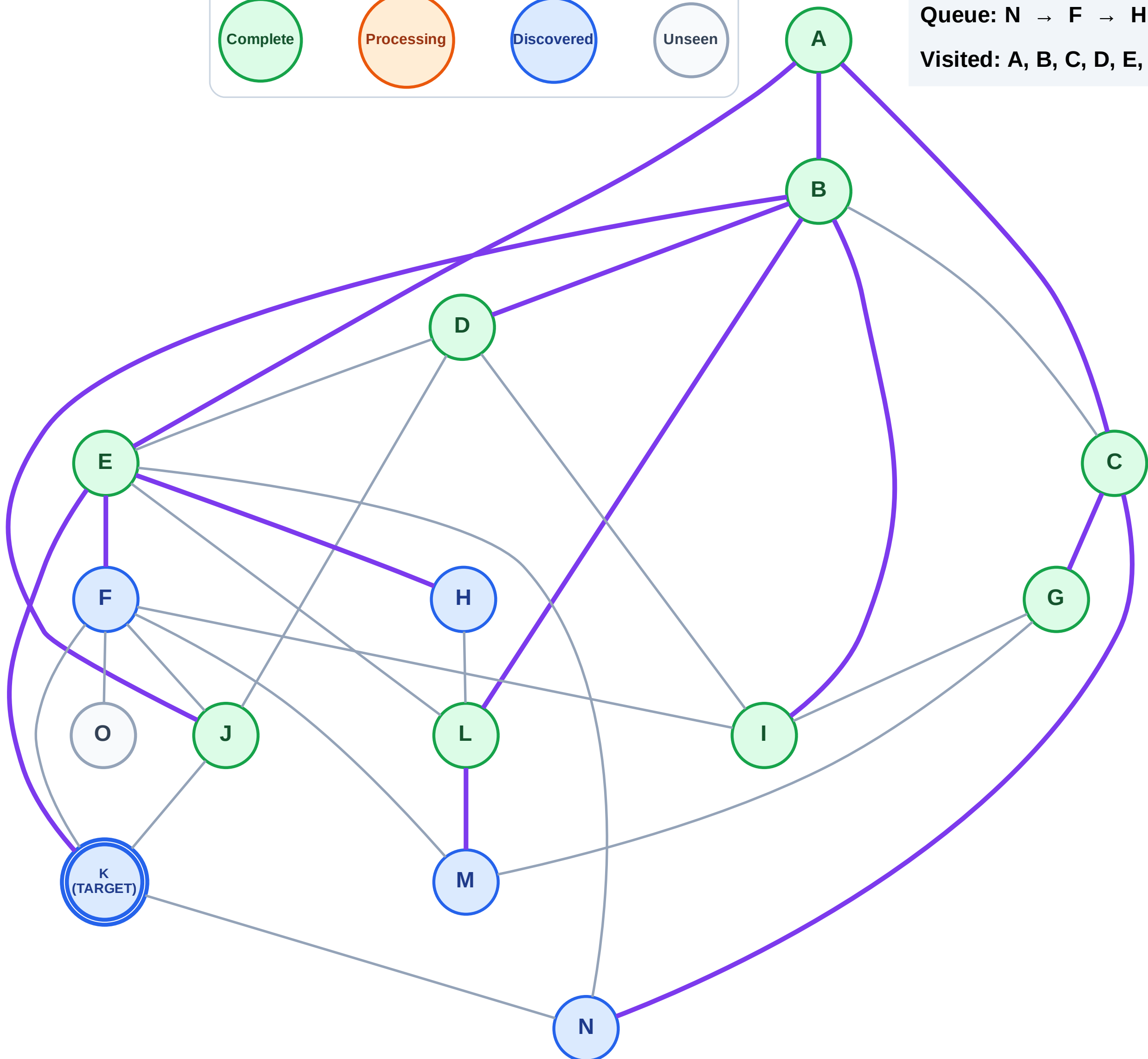
Step 19: No new neighbors from G

## Legend



Queue: N → F → H → K → M

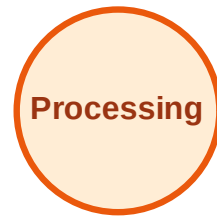
Visited: A, B, C, D, E, G, I, J, L



# BFS Traversal

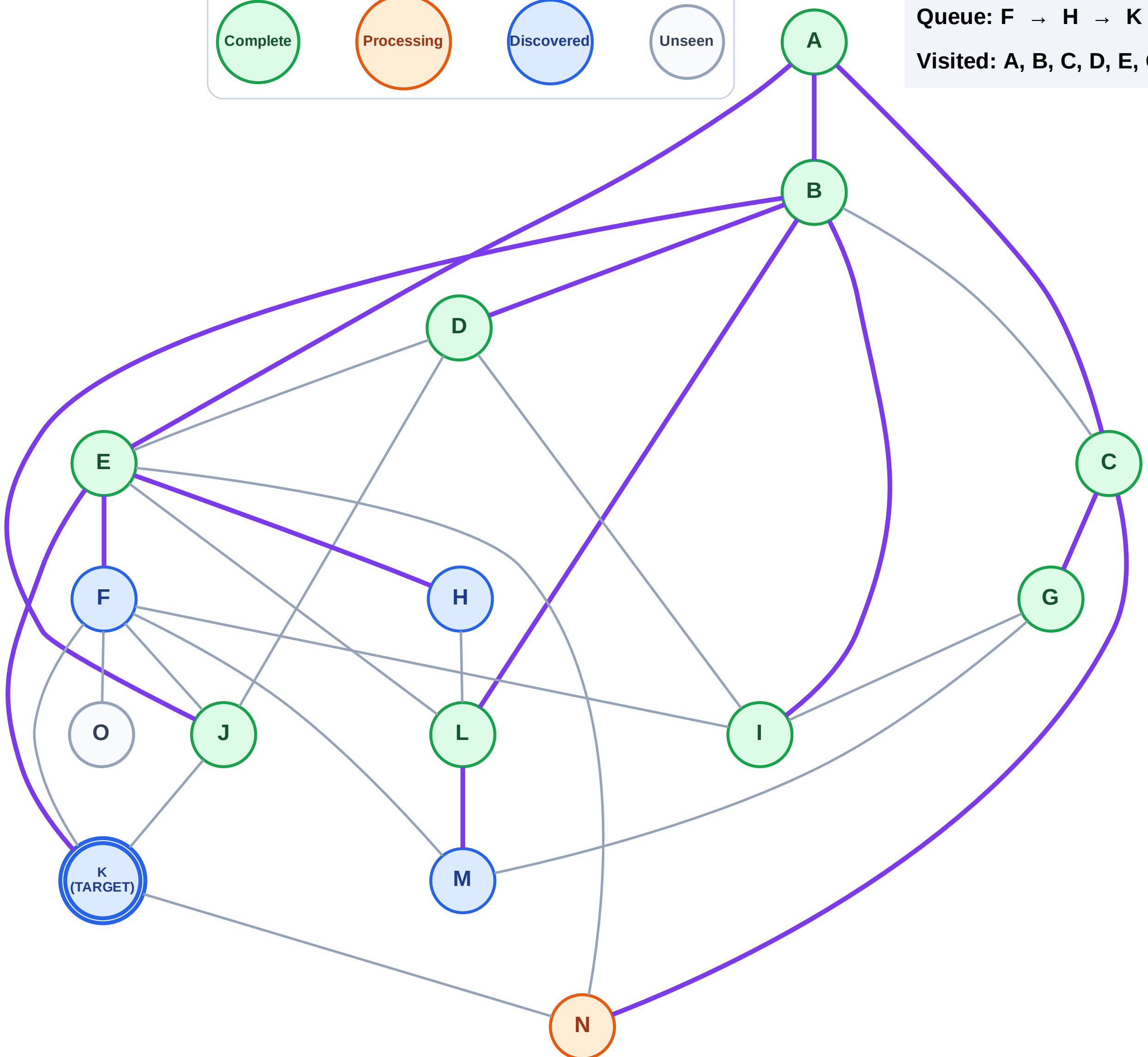
Step 20: Process node N

## Legend



Queue: F → H → K → M

Visited: A, B, C, D, E, G, I, J, L



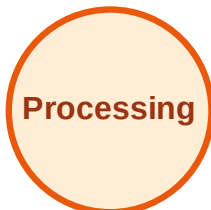
# BFS Traversal

Step 21: No new neighbors from N

## Legend



Complete



Processing



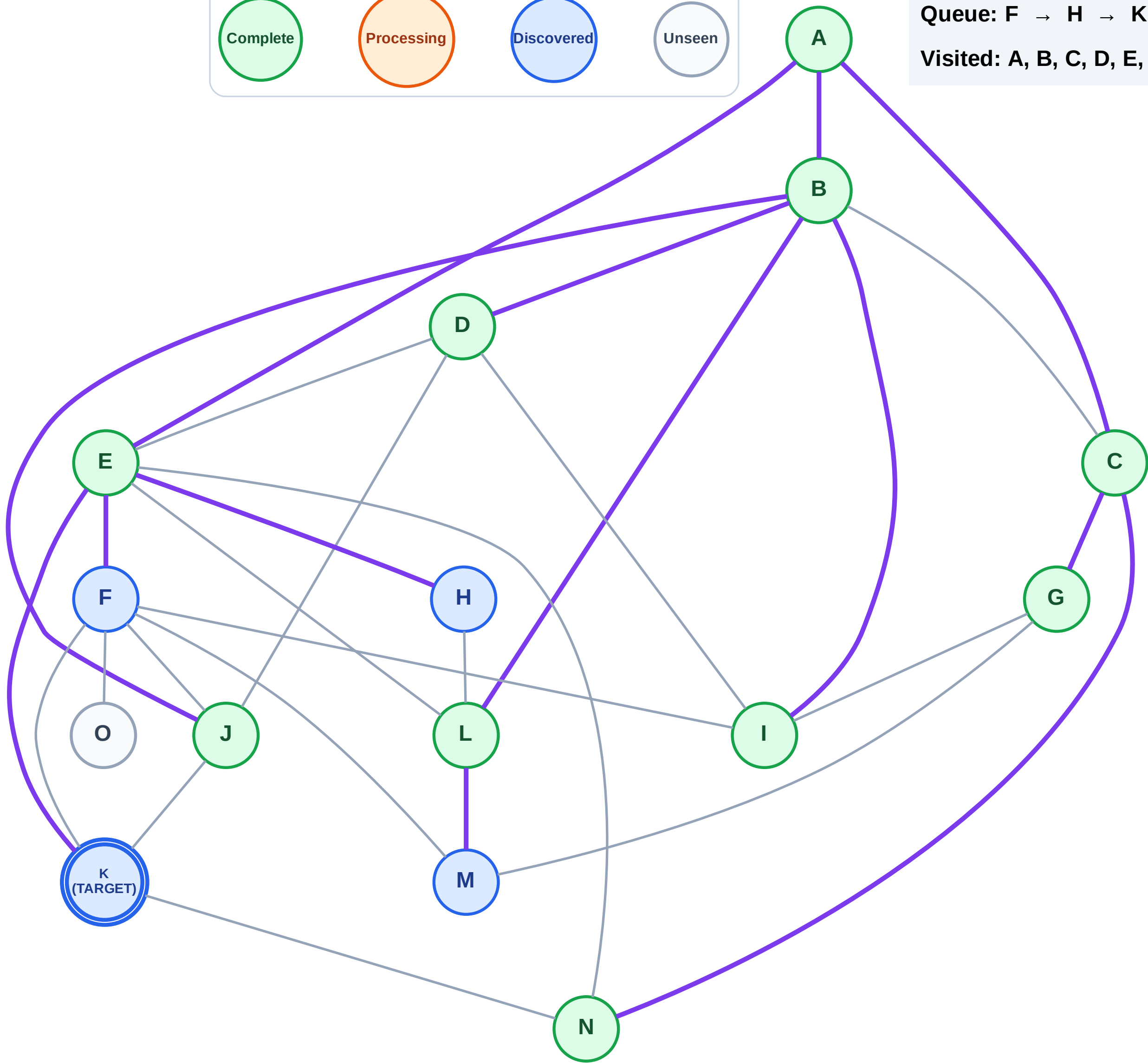
Discovered



Unseen

Queue: F → H → K → M

Visited: A, B, C, D, E, G, I, J, L, N



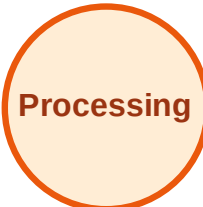
# BFS Traversal

Step 22: Process node F

## Legend



Complete



Processing



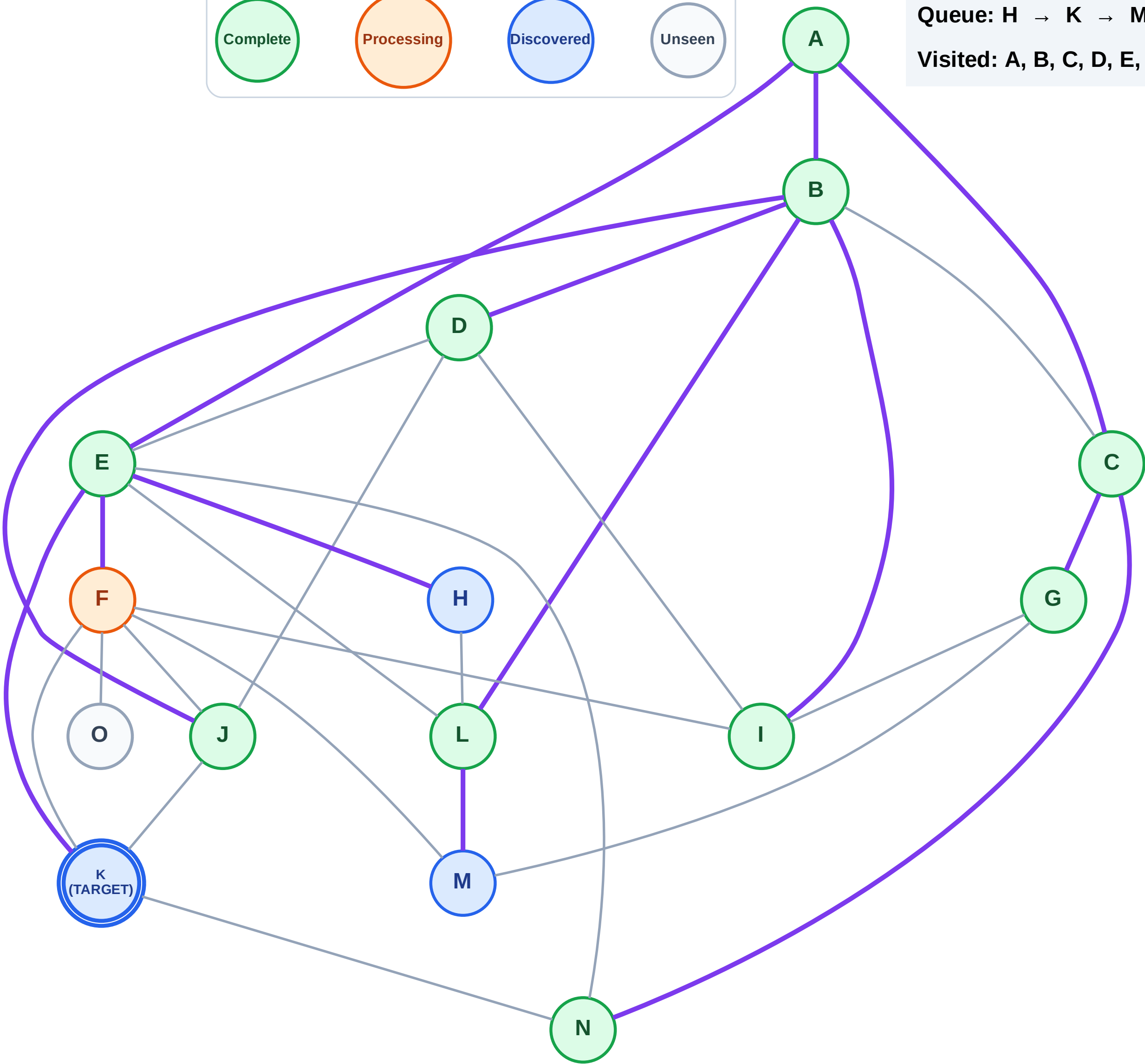
Discovered



Unseen

Queue: H → K → M

Visited: A, B, C, D, E, G, I, J, L, N



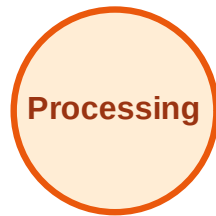
# BFS Traversal

Step 23: Discover O from F

## Legend



Complete



Processing



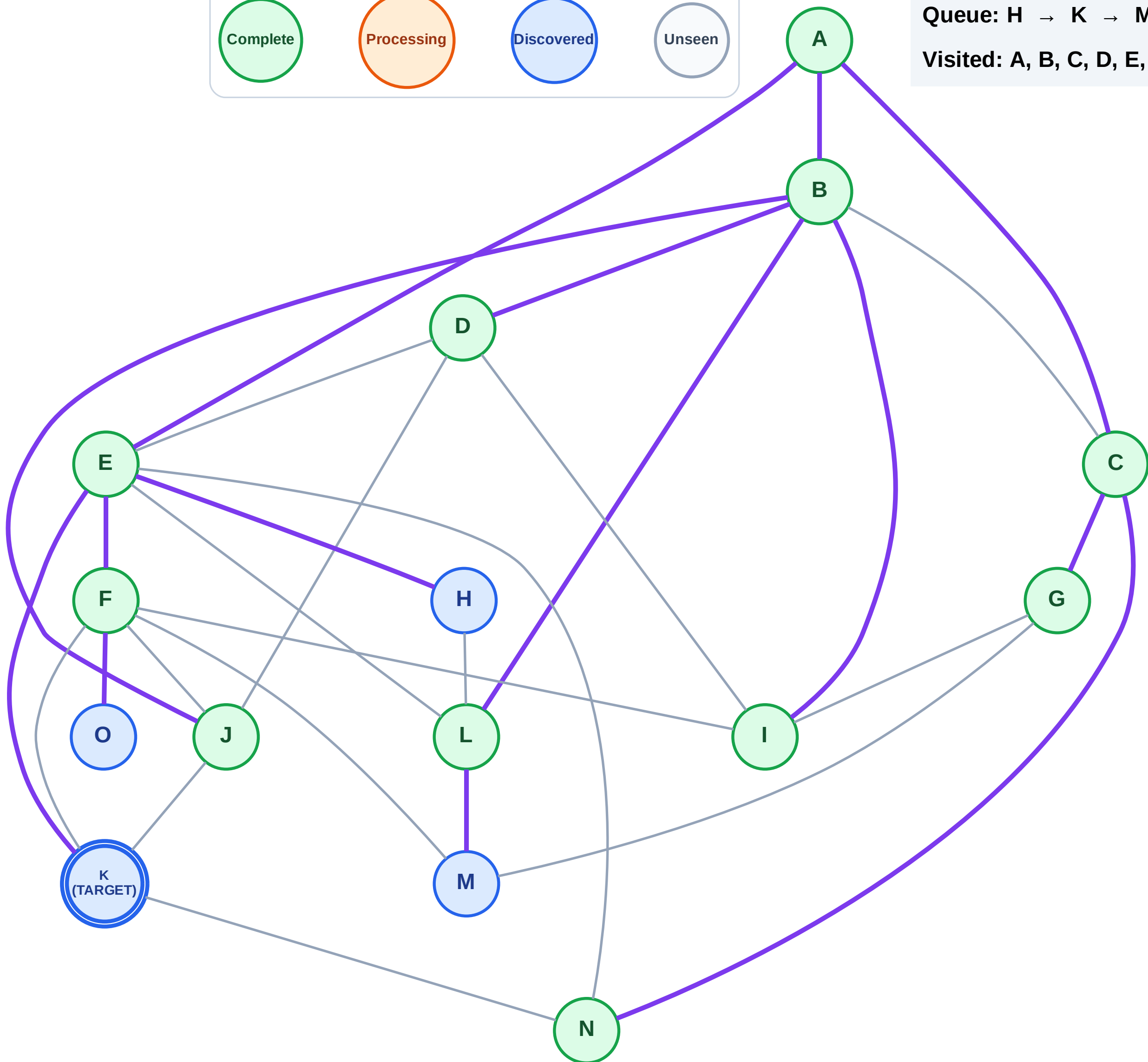
Discovered



Unseen

Queue: H → K → M → O

Visited: A, B, C, D, E, F, G, I, J, L, N



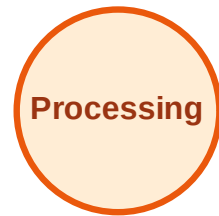
# BFS Traversal

Step 24: Process node H

## Legend



Complete



Processing



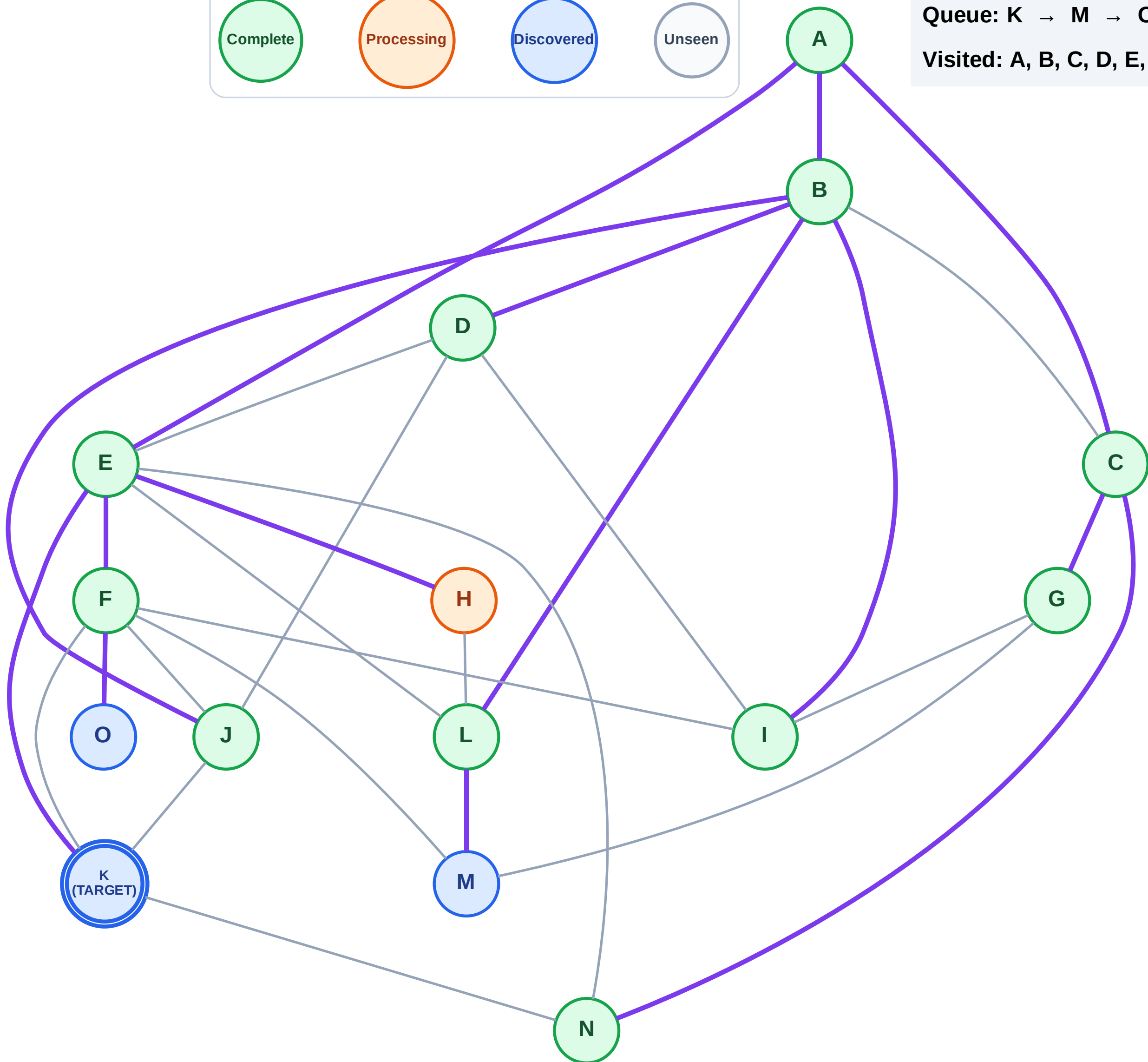
Discovered



Unseen

Queue: K → M → O

Visited: A, B, C, D, E, F, G, I, J, L, N



# BFS Traversal

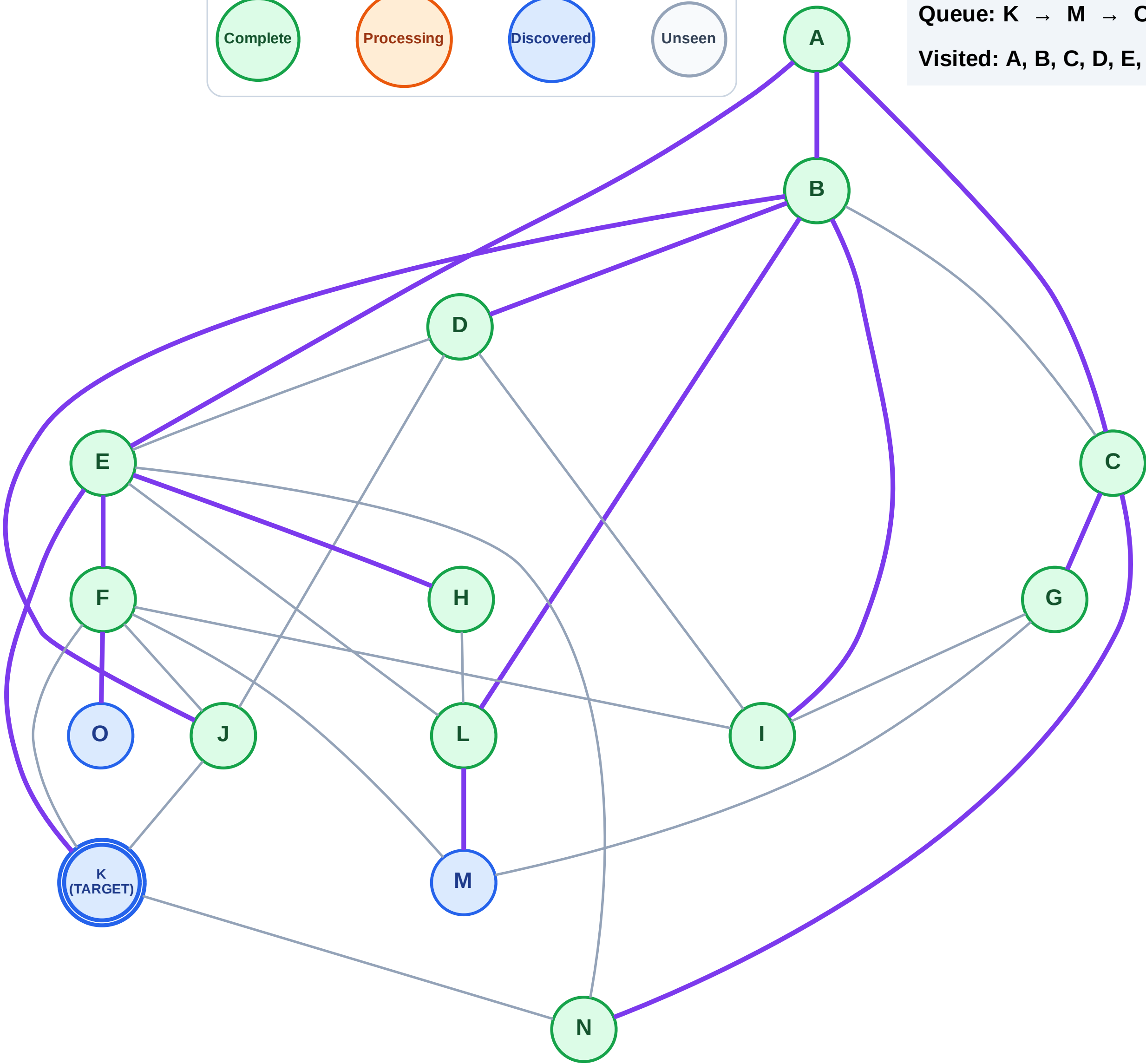
Step 25: No new neighbors from H

## Legend

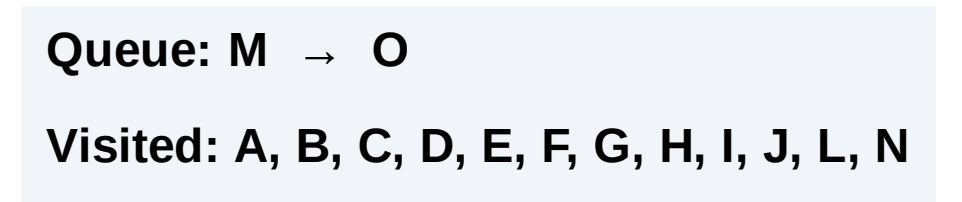


Queue: K → M → O

Visited: A, B, C, D, E, F, G, H, I, J, L, N



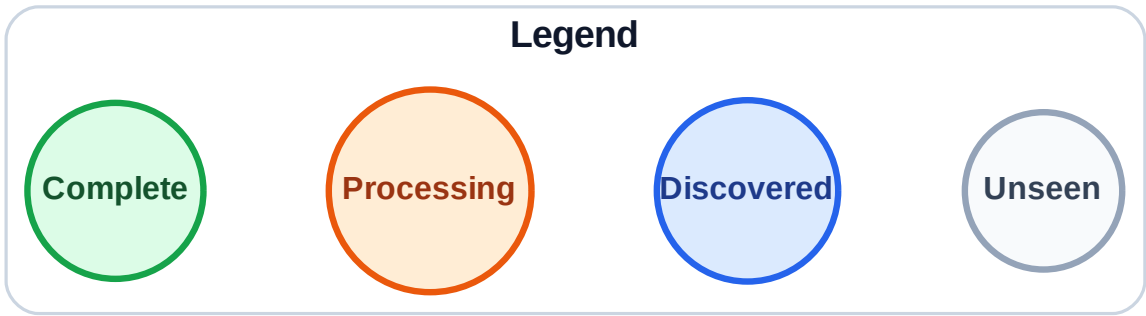
### Step 26: Process node K



# BFS Traversal

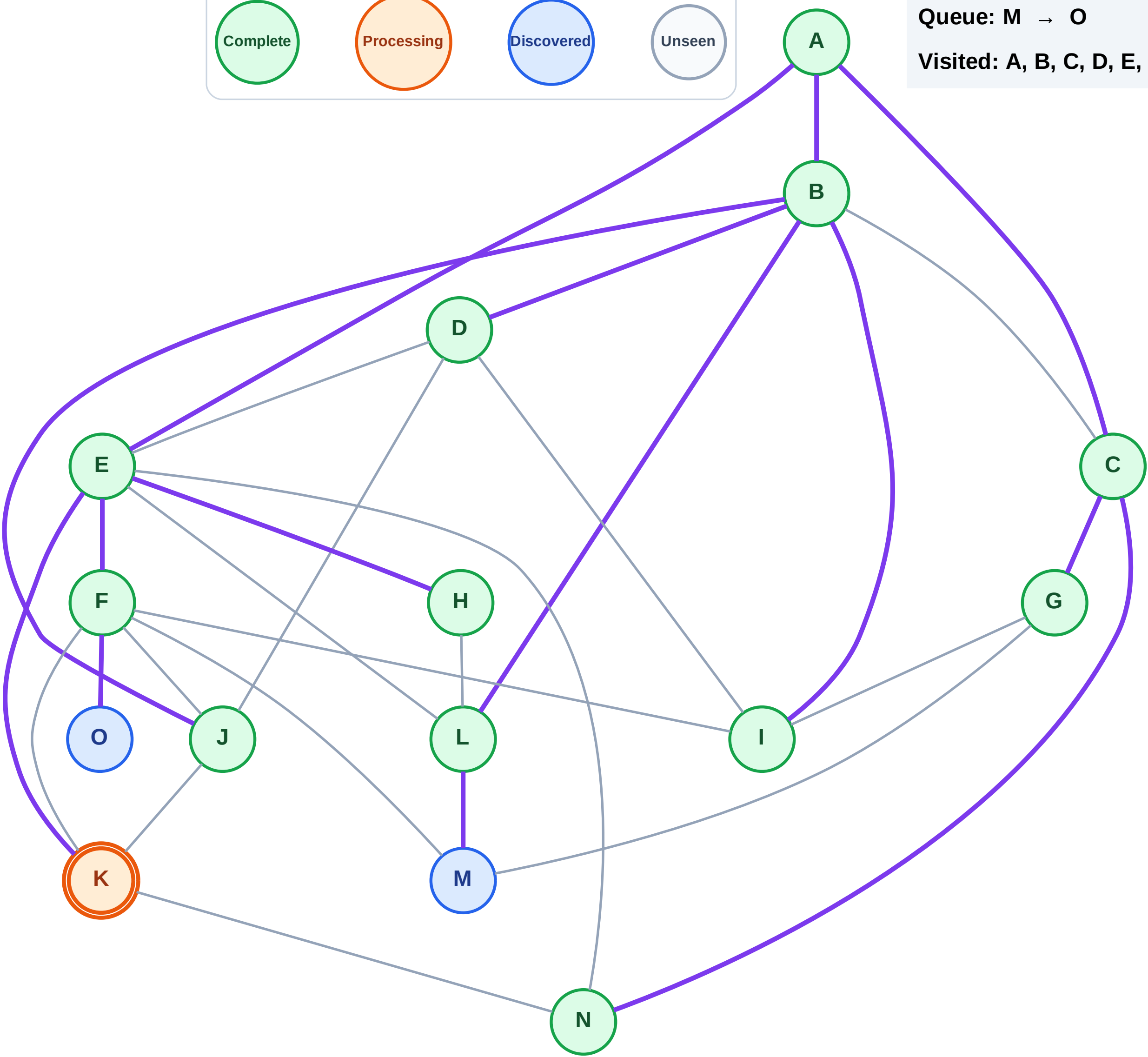
Step 27: TARGET NODE K REACHED! Path (2 hops): A → E → K

## Legend



Queue: M → O

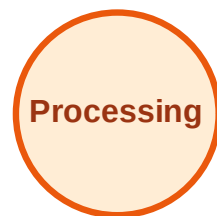
Visited: A, B, C, D, E, F, G, H, I, J, K, L, N



# DFS Traversal

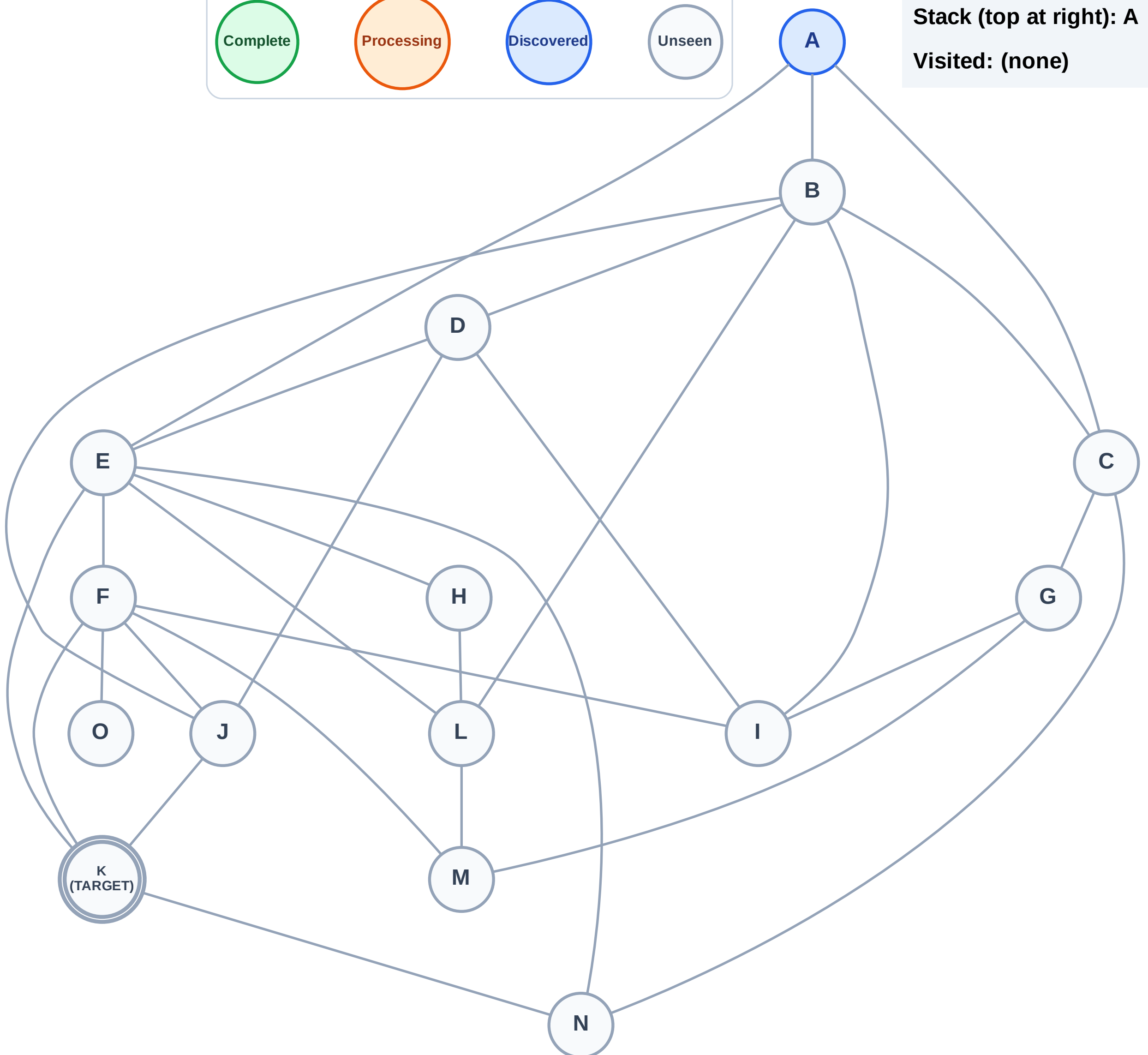
Step 1: Start at A (Seeking target node K)

## Legend



Stack (top at right): A

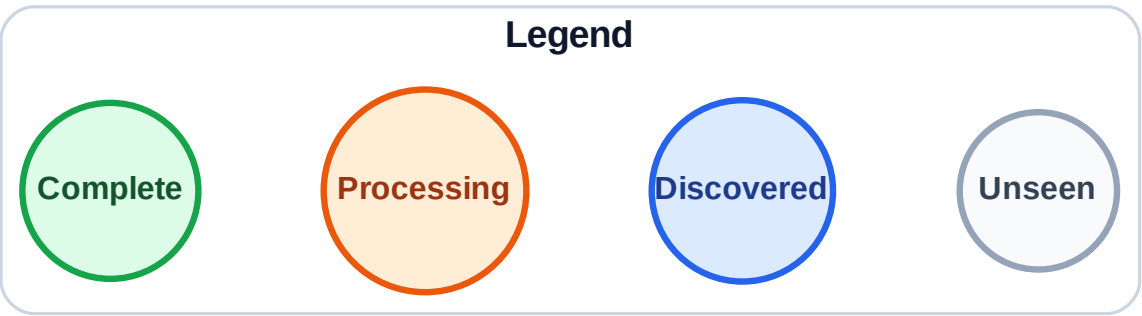
Visited: (none)



# DFS Traversal

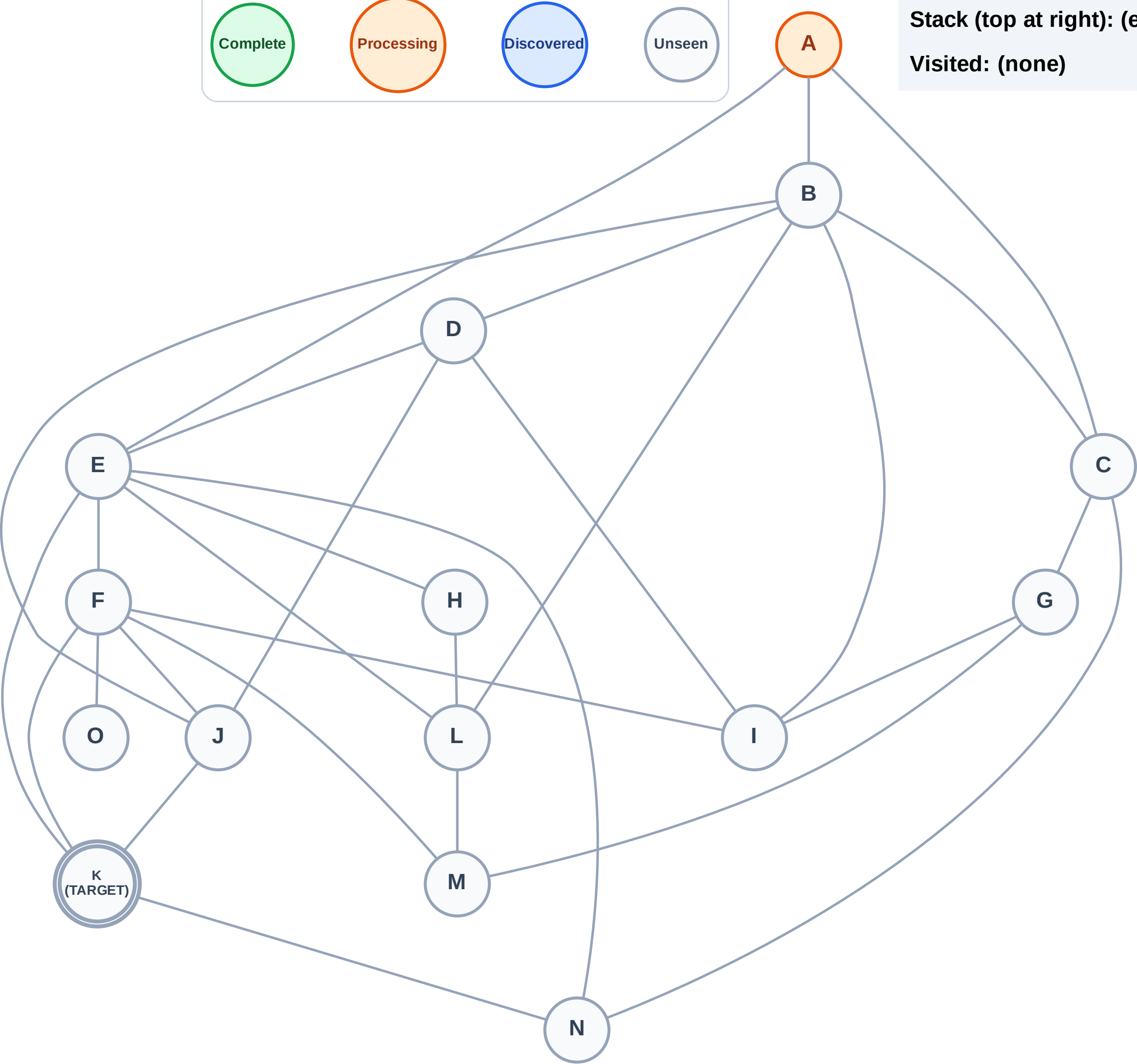
Step 2: Process node A

## Legend



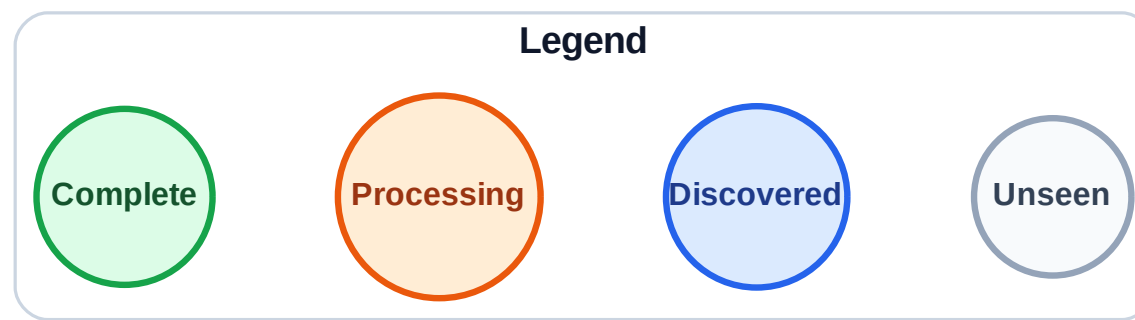
Stack (top at right): (empty)

Visited: (none)



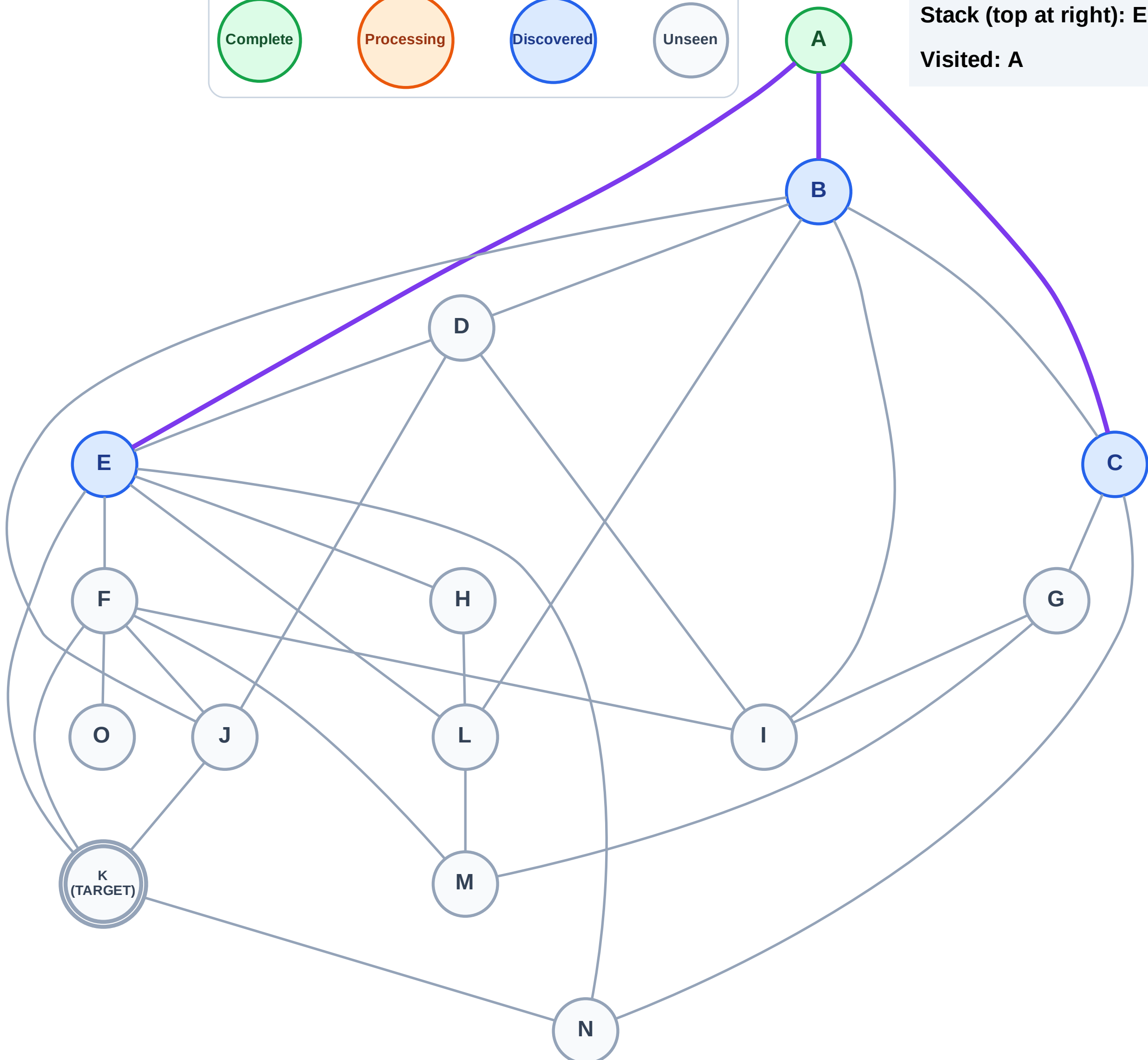
### Step 3: Discover E, C, B from A

### Step 3: Discover E, C, B from A



**Stack (top at right): E | C | B**

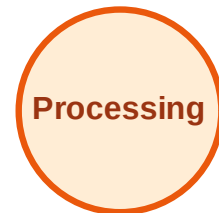
**Visited: A**



# DFS Traversal

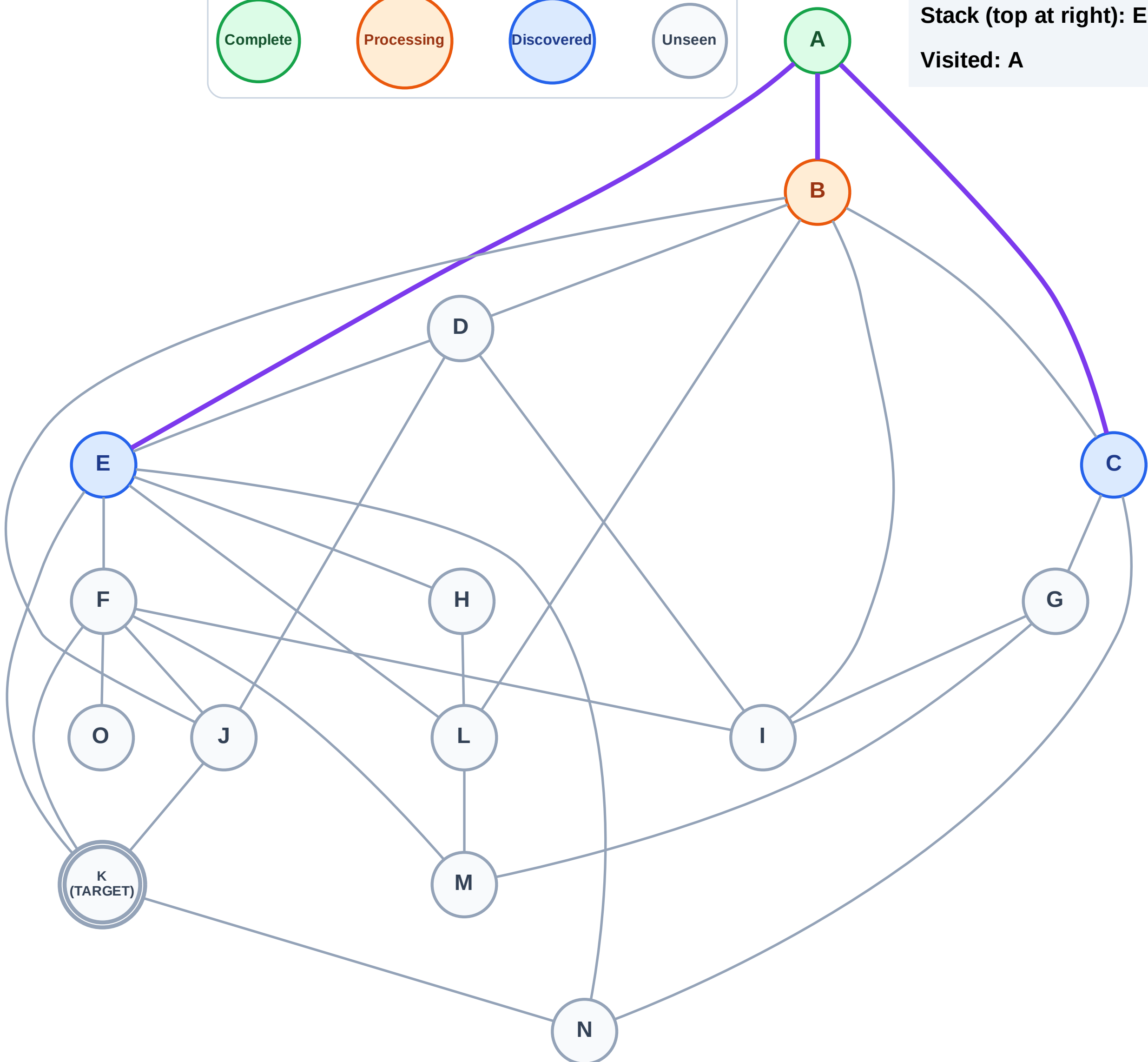
Step 4: Process node B

## Legend



Stack (top at right): E | C

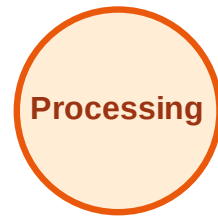
Visited: A



# DFS Traversal

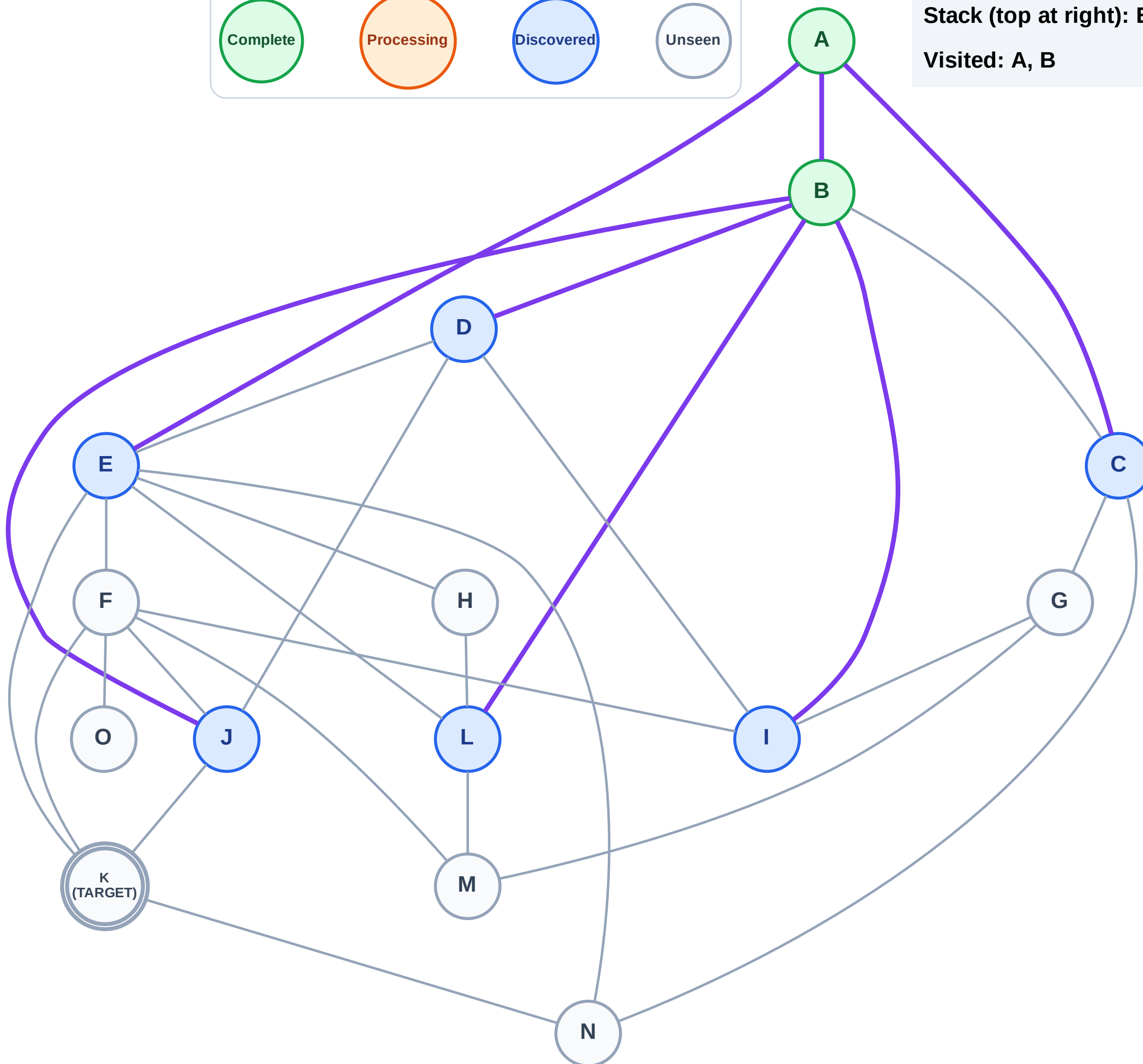
Step 5: Discover L, J, I, D from B

## Legend



Stack (top at right): E | C | L | J | I | D

Visited: A, B



# DFS Traversal

Step 6: Process node D

## Legend



Complete



Processing



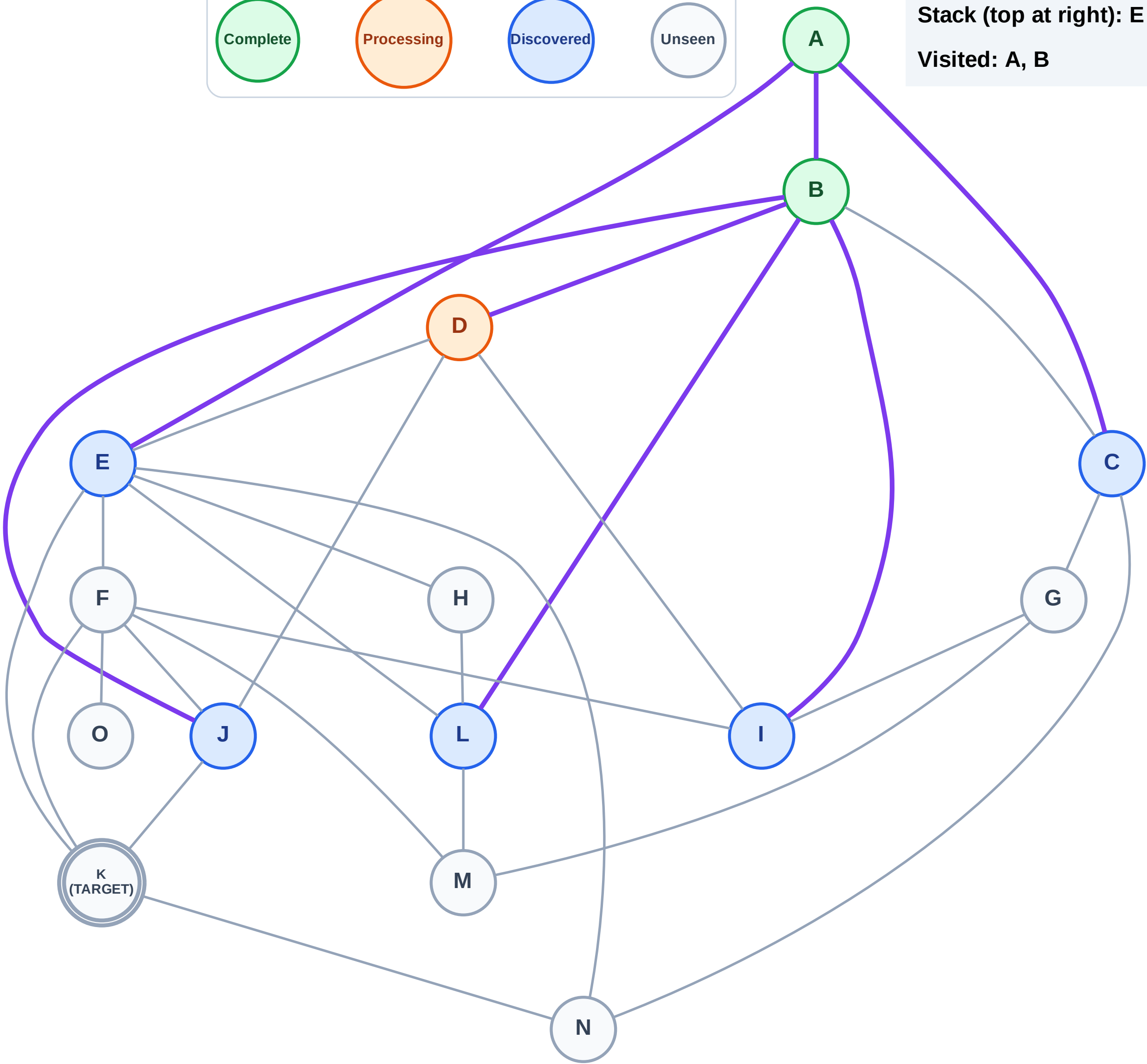
Discovered



Unseen

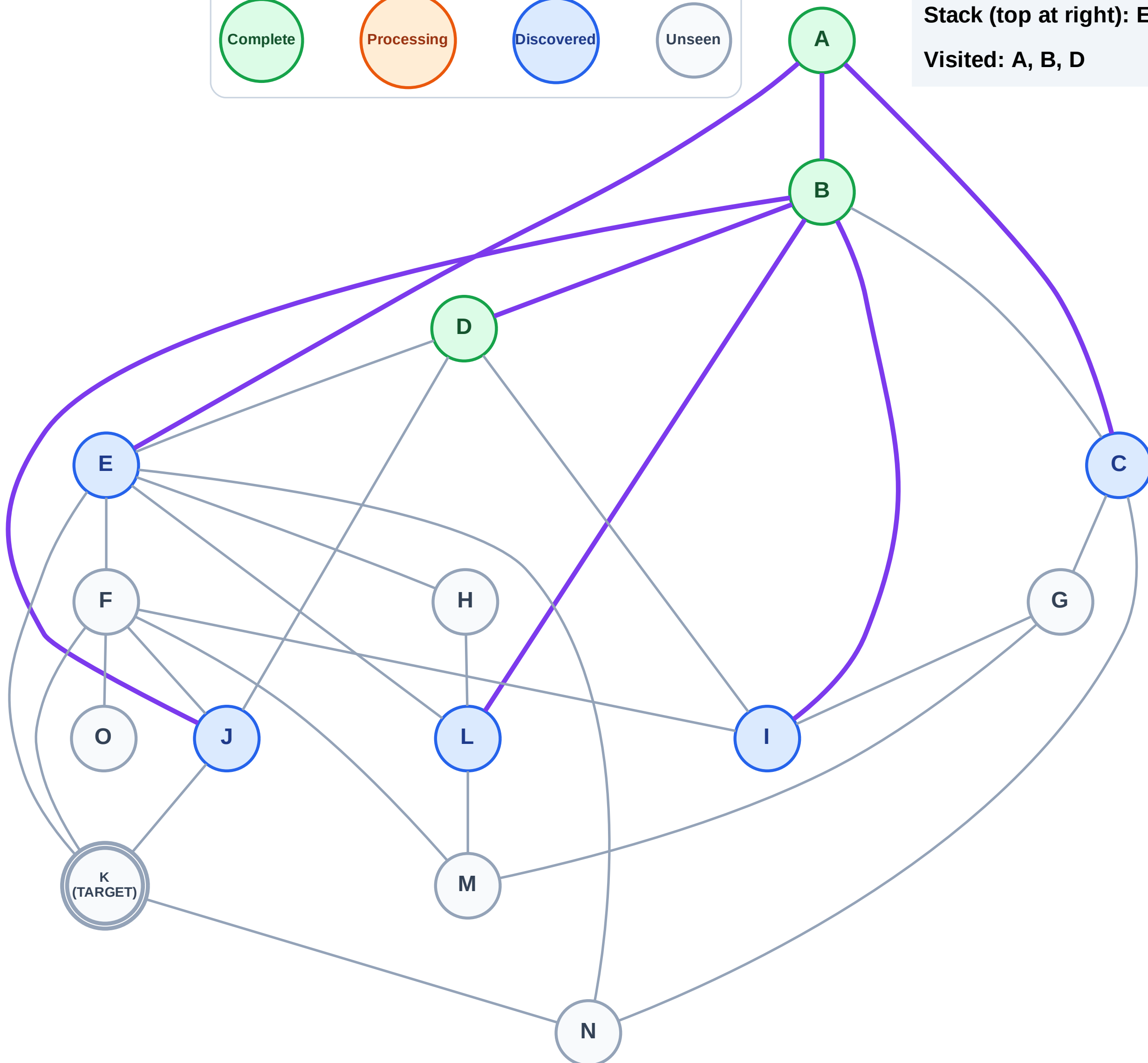
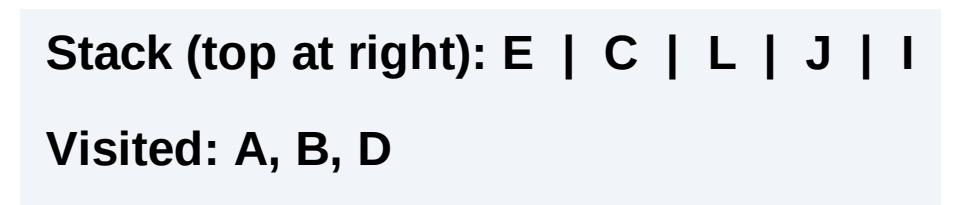
Stack (top at right): E | C | L | J | I

Visited: A, B



Step 7: No new neighbors from D

Step 7: No new neighbors from D



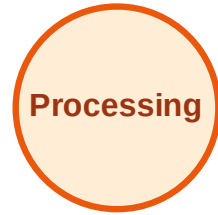
# DFS Traversal

Step 8: Process node I

## Legend



Complete



Processing



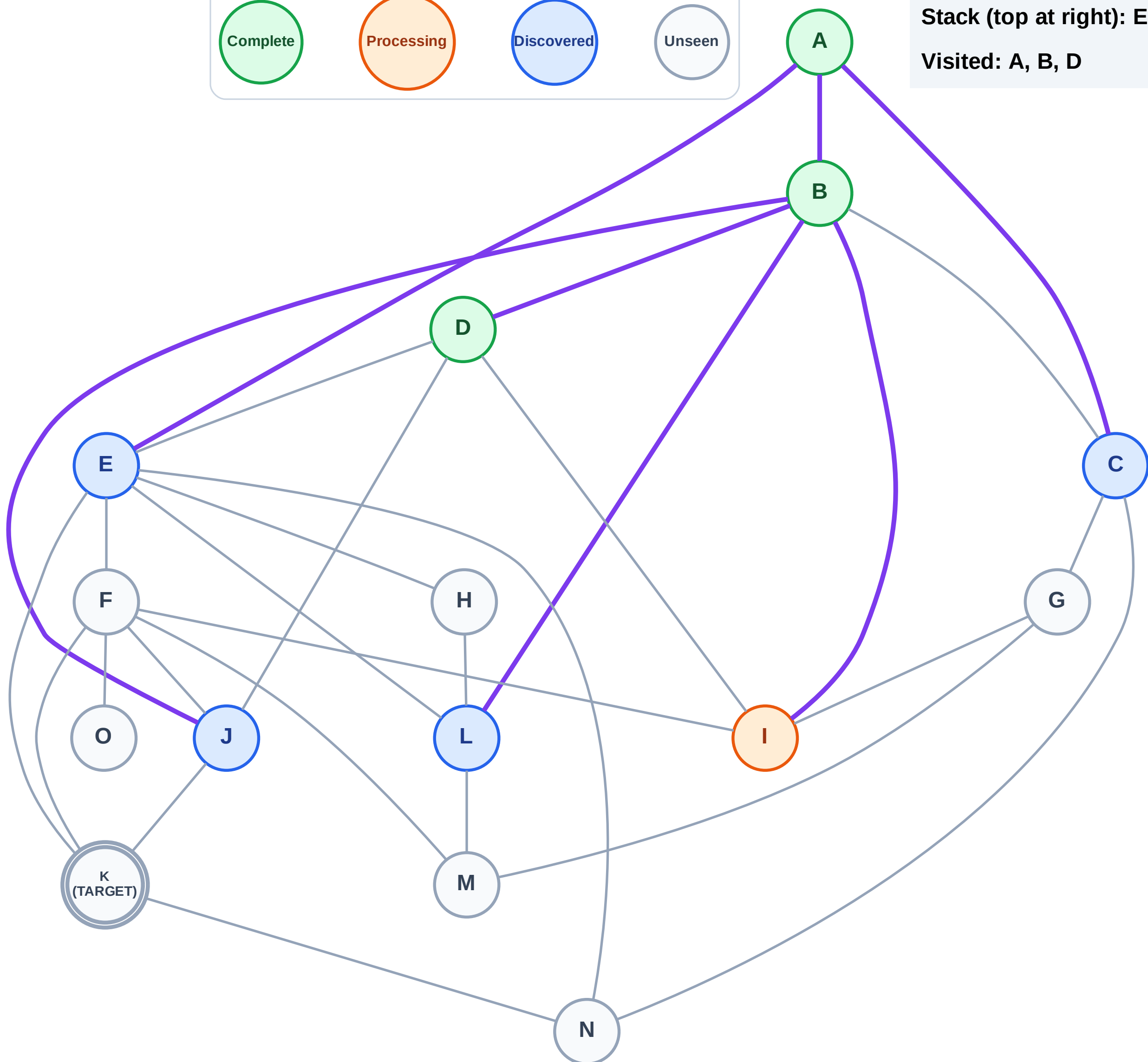
Discovered



Unseen

Stack (top at right): E | C | L | J

Visited: A, B, D



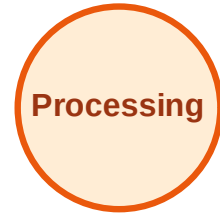
# DFS Traversal

Step 9: Discover G, F from I

## Legend



Complete



Processing



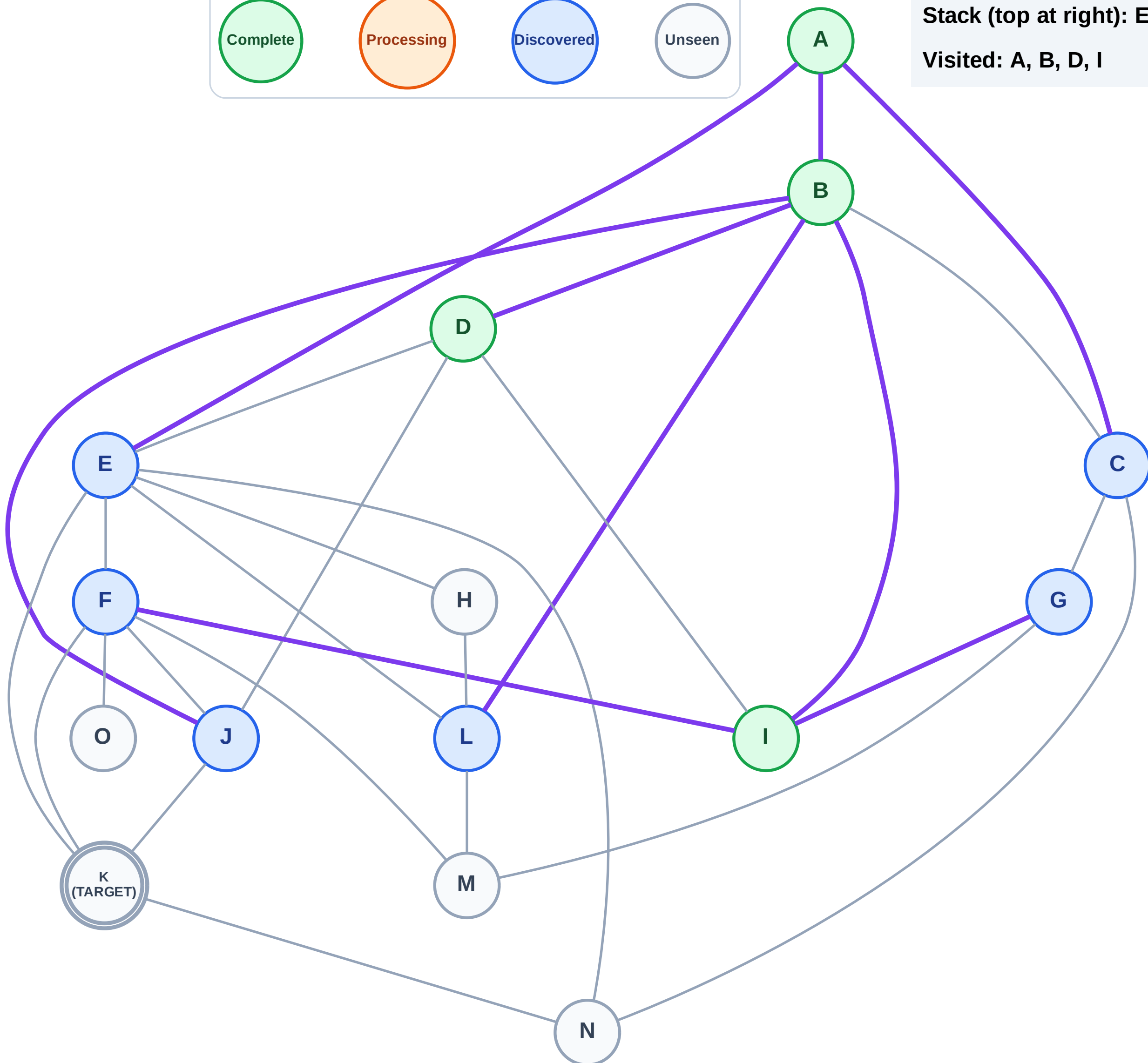
Discovered



Unseen

Stack (top at right): E | C | L | J | G | F

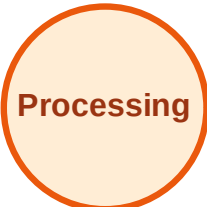
Visited: A, B, D, I



# DFS Traversal

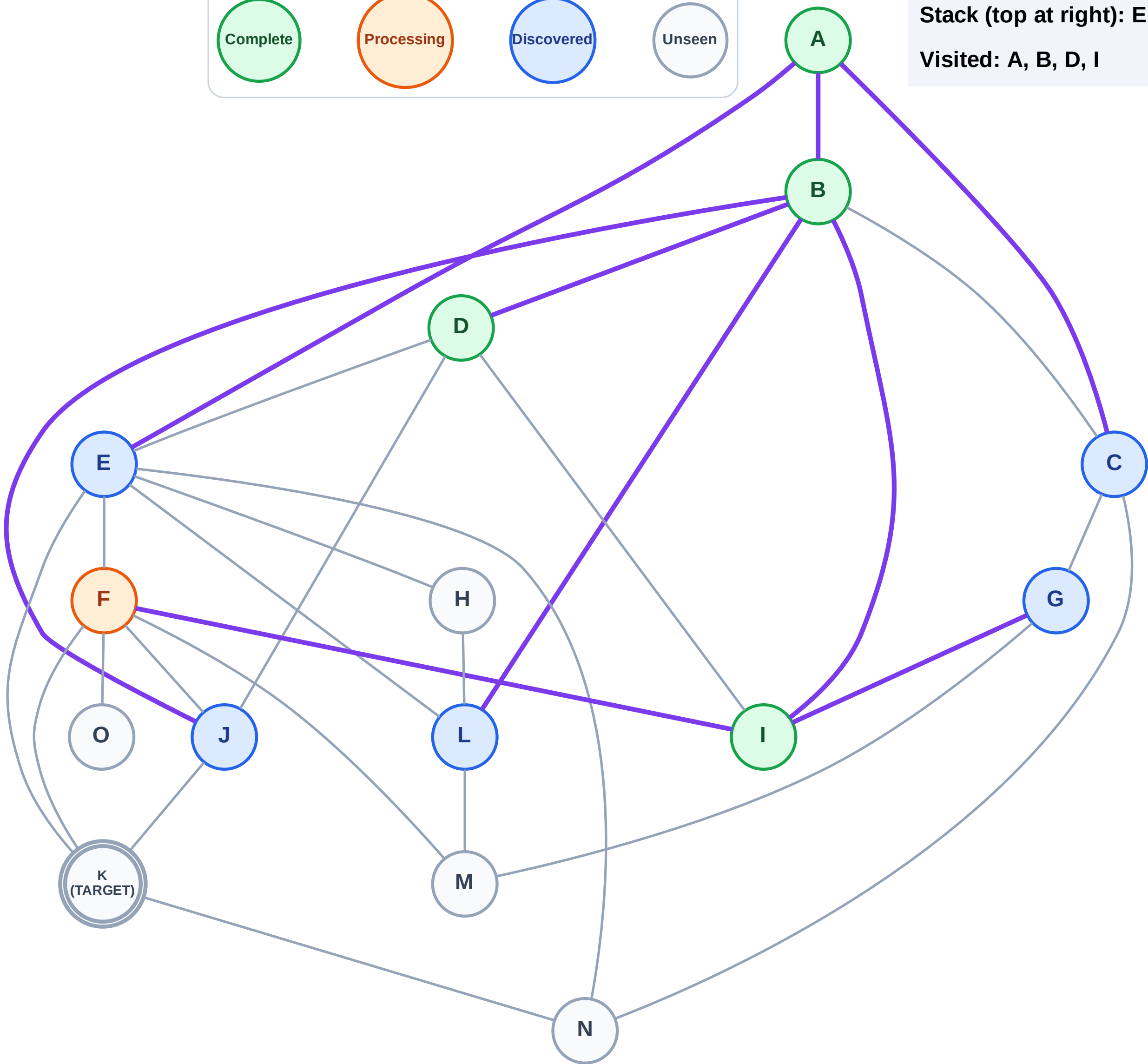
Step 10: Process node F

## Legend



Stack (top at right): E | C | L | J | G

Visited: A, B, D, I



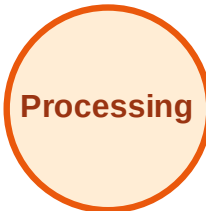
# DFS Traversal

Step 11: Discover O, M, K from F

## Legend



Complete



Processing



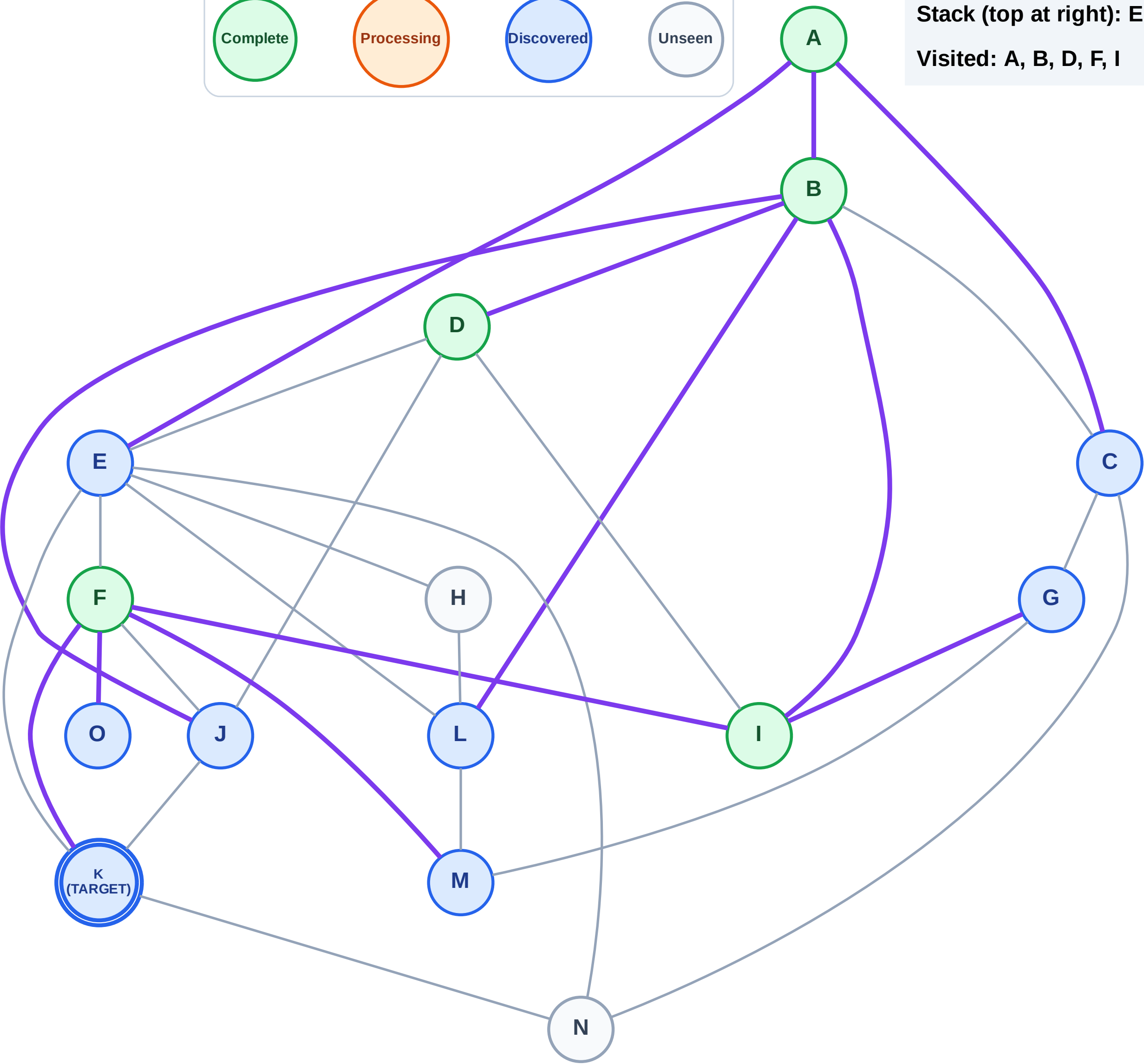
Discovered



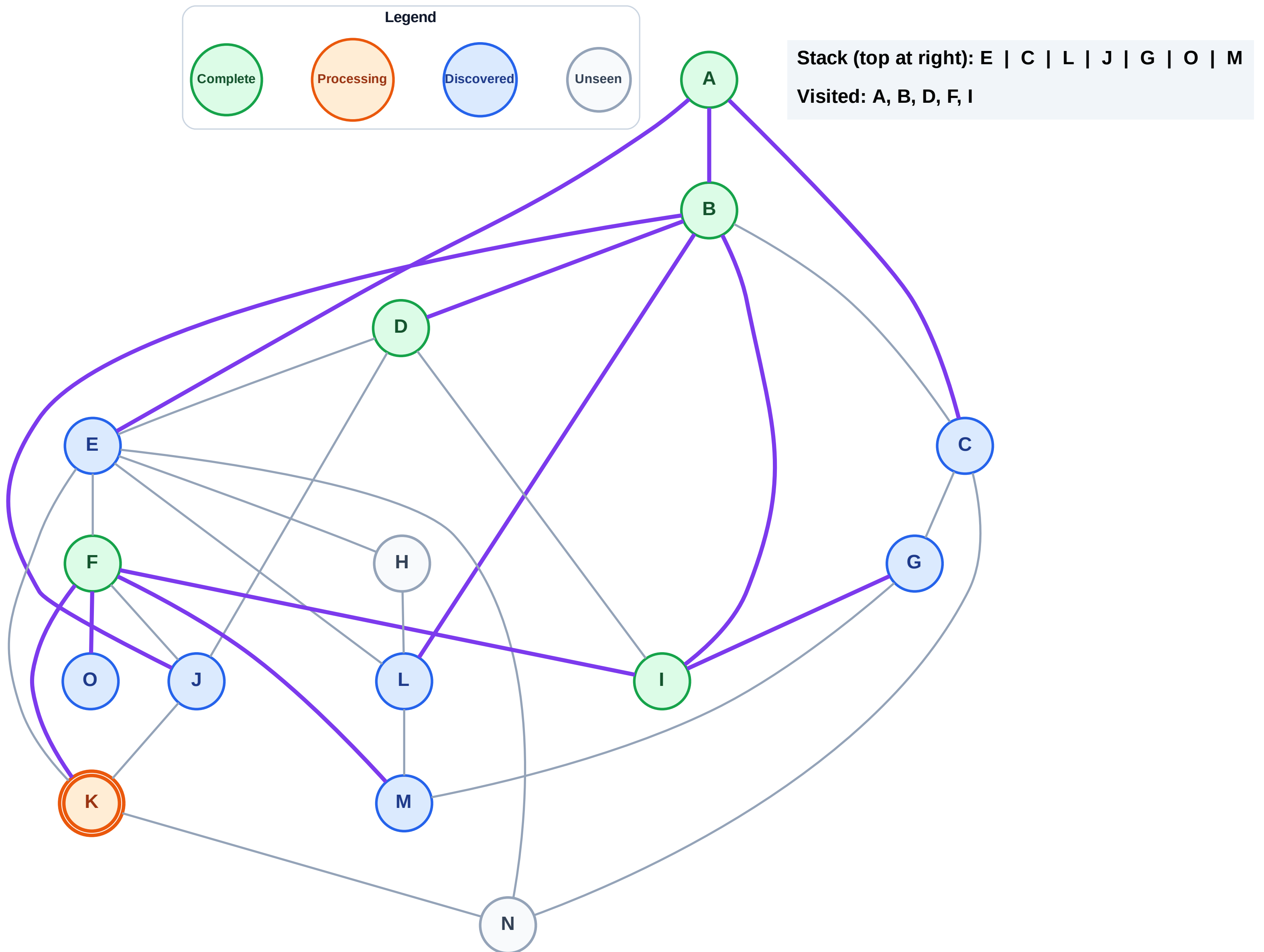
Unseen

Stack (top at right): E | C | L | J | G | O | M | K

Visited: A, B, D, F, I



### Step 12: Process node K



# DFS Traversal

Step 13: TARGET NODE K REACHED! Path (4 hops): A → B → I → F → K

## Legend



Stack (top at right): E | C | L | J | G | O | M

Visited: A, B, D, F, I, K

